MILITARY SPECIFICATION

MULTIPLEXER, TD-1237(P)/G

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of the Multiplexer, TD-1237(P)/G. This unit is also known as the Master Group Multiplexer (MGM) and is one of the electronic units that comprise the Digital Group Multiplex (DGM) family of cable system equipment developed by the Joint Tactical Command, Control and Communications Agency (JTC^3A, formerly TRI-TAC). See 6.1 for intended use and functional description of the unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this doucment should be addressed to Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 5820

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

SPECIFICATION

Military

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-2073-2	Packaging Requirements Code
MIL-STD-45662	Calibration Systems Requirements

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments

DRAWINGS

Communications-Electronics Command

DL-SM-B-875931	Multiplexer, TD-1237/G
SM-D-875948	Circuit Card Assembly(CCA) BITE 1
SM-D-875955	CCA, Mastergroup Receive
SM-D-875959	CCA, Format Generator
SM-D-875961	CCA, Framing
SM-D-875962	CCA, Orderwire Mux/Demux
SM-D-876007	CCA, Scanner
SM-D-876010	CCA, Receive Rate Generator
SM-D-884389	CCA, Group Rate Selector Assembly
SM-D-884492	CCA, Group Logic A
SM-D-884567	CCA, Multi Vdc
SM-D-884579	CCA, AC Input
SM-D-884593	CCA, 5 Vdc
SM-D-959320	CCA, Group Logic B

(Copies of specifications, standards, handbooks, drawings, and publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with DL-SM-B-875931, and the requirements specified herein.

3.2.2 Printed wiring. The printed wiring shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirements 30 and 64.

3.3 Parts, materials, and processes - general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-P-14072 and the equipment drawing DL-SM-B-875931.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. (See 4.15) A unique serial number is required on each nomenclatured item and CCA. The numbers shall be consecutively assigned.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled TD-1237/G shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in and (2) the successful completion of the operational test of 4.7.8 after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Balanced equipment-side interface signal characteristics. (See 4.7.1) The multiplexer accepts on its group side 12 balanced Non-Return-to-Zero (NRZ) group data and squarewave timing signals at bit rates from 72 to 4915.2 kb/s plus two 16 kb/s channels which can be used for either orderwire or telemetry. The groups are asynchronously multiplexed into the mastergroup at bit rates of 9.36 or 18.72 Mb/s, selectable. The 16 kb/s channels are synchronously multiplexed into the master groups. The mastergroup output is balanced NRZ data and timing. In the opposite direction, a mastergroup is accepted in the same form, demultiplexed, and 12 groups are supplied in the same form on the group side. The two 16 kb/s channel outputs are also balanced NRZ data and squarewave timing. The multiplexer accepts a station clock signal as a balanced squarewave. The frequency of the clock shall be 16 or 32 kHz, selectable.

3.7.1.1 Line-to-line voltage levels and signal balance. (See 4.7.1.1) The line-to-line voltage shall be between +0.1 and +4.0 volts for a digital 1 and between -0.1 and -4.0 for a digital 0, measured into a 100 ohm dummy load. The magnitude of the difference between the magnitudes of the digital 1 level and the digital 0 level shall be less than 10% of the larger of the two.

3.7.1.2 Line-to-ground voltage. (See 4.7.1.2) The magnitude of the line-to-ground voltage shall not exceed 3 volts.

3.7.1.3 Clock/data phasing. (See 4.7.1.3) For those data signals having a related clock signal, the time difference between the upward transition of the clock signal and either upward or downward transition of the data signal shall be no more than $\pm$10 ns for the mastergroup signals, 29 ns for group rate signals, or 1500 ns for 16 kb/s signals. Times shall be measured at the zero voltage point of the signals.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.7.1.4 Offset. (See 4.7.1.4) The magnitude of the offset voltage measured between the center tap of a 100-ohm dummy load and ground shall not be greater than 3 volts.

3.7.1.5 Slew rate. (See 4.7.1.5) When 100 feet of RG-108A/U cable terminated in a 78-ohm dummy load is connected to the driver output, the transition between ±40mV (either direction) shall take no more than 10 ns for group rate signals and 100 ns for the 16 kb/s signals.

3.7.1.6 Receiver dynamic range. (See 4.7.1.6) The receiver shall deliver the proper digital output when a line-to-line voltage of +50 mV to 4V (for a digital 1) or -50 mV to -4V (for a digital 0) is applied to the receiver input terminals.

3.7.1.7 Input load impedance. (See 4.7.1.7) The input impedance shall be 100 ohms ±10%.

3.7.1.8 Clock asymmetry. (See 4.7.1.8) The positive half of the clock waveform shall be equal to half the period ±6 ns for the mastergroup timing signals, ±15 ns for group rate timing signals, and ±1500 ns for 16 kb/s timing signals.

3.7.2 Multiplexing. (See 4.7.2) The unit shall be capable of multiplexing up to 12 groups of bit rates of 72, 128, 144, 256, 288, 512, 576, 1024, 1152, 1536, 2048, 2304, 4096, 4608, and 4915.2 kb/s signals in any combination, with a maximum total input bit rate of 18,432 kb/s for the 18,720 kb/s mastergroup output bit rate. The unit shall be capable of multiplexing 72, 128, 144, 256, 288, 512, 576, 1024, 1152, 1536, 2048, and 2304 kb/s signals in any combination with a maximum total input bit rate of 9216 kb/s for the 9360 kb/s mastergroup output bit rate. The equipment shall accept all permutations of the above combinations which comply with table I. The groups shall be multiplexed without loss of Bit Count Integrity (BCI) when they are timed from clocks that are asynchronous to each other and to the mastergroup. The multiplexer shall accept clock frequency errors of up to ±30 ppm from nominal. The unit shall be capable of multiplexing up to two channels at 16 kb/s. These channels shall be synchronous to the station clock input.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE I. TD-1237 group rate restrictions.

Mastergroup Rate, Mb/s	Group Rate, kb/s	Do not apply to groups numbered
18.72	4915.2	7, 8, 9, 10, 11, 12
18.72	4608	8, 9, 10, 11, 12
18.72	4096	8, 9, 10, 11, 12
9.36	2304	8, 9, 10, 11, 12
9.36	2048	9, 10, 11, 12
9.36	1536	11, 12

3.7.3 Timing modes. (See 4.7.3) The following timing modes shall be available in the equipment.

3.7.3.1 Operation from internal timing. (See 4.7.3.1) The equipment shall operate from the internal oscillator when the timing select switch is set to station clock and the station clock is absent. A loss of input (LOI) alarm shall be generated in this event. The internal oscillator shall have an accuracy of ±5 ppm or better.

3.7.3.2 Operation from receive mastergroup timing. (See 4.7.3.2) In this mode, transmit timing is derived from the received mastergroup signal by the Cable Driver Modem MD-1024/G or radio.

3.7.3.3 Operation from station clock. (See 4.7.3.3) In this mode transmit timing is obtained from a 16kHz or 32 kHz external station clock.

3.7.4 Framing. (See 4.7.4) The frame pattern and frame rate shall enable the equipment to acquire frame synchronization within 20 milliseconds, with a minimum probability of 0.9, and maintain frame synchronization for not less than 24 hours, with a minimum probability of 0.9, in an error environment of 0.1% BER random errors. Loss of frame shall be detected within 100 ms with a minimum probability of 0.9.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.7.5 Prime power. (See 4.7.5) The ac primary power shall be 115V ±10% at frequencies of 50, 60, and 400 Hz ±5%, single phase. The power input to the equipment shall not exceed 200W. The ac primary shall conform to MIL-STD-1332 for Type 1, Class 1 and 2, Mode I, II, and III generator sets.

3.7.6 Built-in test equipment (BITE) and alarms. (See 4.7.6) The built-in test equipment (BITE) shall monitor signals in and out of the unit and internal circuits. Go-no-go fault location circuits shall isolate failures to a particular printed circuit card or lowest replaceable unit (LRU). Indicating devices shall be provided to insure proper operation of the equipment and to monitor the dc voltages, go-no-go fault isolation, and other critical internal and external signals. The BITE shall operate without interrupting traffic or interfering with normal system operation. Removal of the BITE card shall not affect operation of the unit. Failure alarms shall be provided for channel failure, common power supply failure, and inter-unit signal malfunction.

3.7.6.1 BITE alarms. Failures detected by the BITE circuitry cause relay contact closures accessible externally at a connector on the rear panel. The outputs described below shall be available.

a. S alarm. The S alarm shall indicate any failure detected including loss of input power. This shall also provide a visible indication on the front panel, except for loss of input power.

b. SA alarm. The SA alarm indicates that the malfunction is within the unit. If the power switch is set to OFF, there shall be no SA alarm.

c. FS alarm. The FS alarm indicates that the unit has lost frame.

3.7.7 Operational check. (See 4.7.7)

3.7.7.1 Group data error check. (See 4.7.7.1) There shall be no errors for a period of 10 seconds.

3.7.7.2 Orderwire and telemetry error check. (See 4.7.7.2) There shall be no errors for a period of 30 seconds at 16 kb/s.

3.7.7.3 Alarm monitor check. (See 4.7.7.3) There shall be no alarms generated.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.7.8 Environmental operational test. (See 4.7.8) The equipment shall be tested with the Group Logic B groups operating at 4915.2 kb/s and the Group Logic A groups operating at 576 kb/s.

3.7.8.1 Group data error check. (See 4.7.8.1) There shall be no errors in 10 seconds.

3.7.8.2 Orderwire and telemetry error check. (See 4.7.8.2) There shall be no errors in 30 seconds.

3.7.8.3 Alarm monitor check. (See 4.7.8.3)

a. No alarms shall be generated, that is, there shall be open circuits on the S, SA and FS alarm outputs. (A short indicates alarm).

b. Removal of the Group cable from Group 1 of the auxiliary multiplexer shall cause an S and LOI alarm.

3.7.8.4 Sync/frame check. (See 4.7.8.4)

a. An S and LOI alarm shall be present on the UUT when the auxiliary multiplexer is turned off.

b. The S and LOI alarms on the UUT shall be extinguished when the auxiliary multiplexer is turned on.

3.7.9 Jitter. (See 4.7.9) The maximum frequency deviation of any output group signal from the frequency of the input group signal shall be $\pm$1.25 ppm or less for long term group frequency offsets within $\pm$1 ppm of nominal, and $\pm$2 or less for long term offsets within $\pm$30 ppm of nominal.

3.8 Service conditions. (See 4.8) The operational test of 3.7.12 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +63°C (+145°F), and shall withstand exposure (non-operating) to ambient air temperatures as high as +71°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient air temperatures as low as -31.7°C (-25°F), and shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall withstand, in both operating and non-operating conditions, the effects of dust particles that may accumulate within the shelter assembly or vehicle.

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of a salt fog atmosphere.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Vibration and shock. (See 4.8.7). The equipment shall withstand vibration and shock induced during vehicular transport as part of a mobile assemblage, over all types of roads and cross-country terrain, and shocks encountered during servicing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3.9 Electromagnetic interference (EMI). (See 4.9) The unit shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4:

CE02	CS02	RE02	RS02
CE03	CS06	RE02.1	RS03
CE04			
CE05			

NOTES:

a. RE02.1 is applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 is applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to 60 Hz and 400 Hz power lines.

3.9.1 Bonds and grounds. Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The unit shall have a specified Mean-Time-Between Failure (MTBF), θ_o of 4000 hours when in the normal modes of operation.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements specified on their respective test requirement specification (TRS) drawings, as referenced in the following subassembly drawings:

CCA	DRAWINGS
BITE 1	*SM-D-875948
Mastergroup Receiver	*SM-D-875955
Format Generator	*SM-D-875959
Framing	*SM-D-875961
Orderwire Mux/Demux	*SM-D-875962
Scanner	*SM-D-876007
Receiver Rate Generator Module	*SM-D-876010
Group Rate Selecter Assembly	SM-D-884389
Group Logic A	*SM-D-884492
Multi Vdc	SM-D-884567
AC Input	SM-D-884579
5 Vdc	SM-D-884593
Group Logic B	*SM-D-959320

* Drawing contains automatic test equipment (ATE) acceptance procedures and programs.

3.13 Operational tests. (See 4.7.8) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.7 after all other inspections and tests are completed to determine that the unit is operating prior to packaging. The operational test of Group A satisfies this requirement for those units for which this is the last inspection or test.

3.14 Systems safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1 and 8, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with the requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

3.16 Nuclear survivability. The equipment shall satisfy the nuclear survivability requirements stated in Section 3 of CR-CX-0173-001 with the exception of paragraphs 3.3.1.1 (BLAST) and 3.3.1.2 (THERMAL).

4.QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections

(1) Quality conformance inspections of equipment before fore packaging. (See 4.5)

(2) Packaging inspection. (See 4.14)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 20 first article units of the complete Multiplexer TD-1237/G. The first article quantity shall be as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table II and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo in-process Group A inspection prior to burn-in.

4.3.3 First article inspection data. The first article inspection test plan and test report(s) shall be as required in the contract or purchase order.

NOTES for table II:

a. The table II inspections 1 through 3, in the order shown, shall be performed on all first articles inspection units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by Inspections 5 and 6 specify the order of inspections for the indicated units. An asterisk (*) indicates the use of an auxiliary TD-1237(P)/G unit.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. All first article units shall be subjected to Group A and Group B inspection.

e. A double asterisk (**) indicates a one-time only engineering test to be covered under a report separate from the first article test report.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

TABLE II. First article inspections.

Inspection	Reqt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
1. Inspections by subsidiary documents	N/A	4.4	Inspections to be performed on all units.												
2. Burn-in	3.6	4.6	Inspections to be performed on all units.												
3. Group A Inspection	(See table III)		Inspections to be performed on all units.												
4. Group B Inspection	(See table IV)		Inspections to be performed on all units.												
5. Group C Inspection															
Subgroup 1															
High Temperature	3.8.1.1	4.8.1.1	1	*											
Low Temperature	3.8.1.2	4.8.1.2	2	*											
Subgroup 2															
EMI	3.9	4.9			1										
Vibration/Shock	3.8.7	4.8.7				1	*								
Subgroup 3															
Humidity (Note c)	3.8.2	4.8.2						1							
Subgroup 4															
Altitude	3.8.3	4.8.3							1		*				
Dust (Note c)	3.8.4	4.8.4							2		*				
Salt Fog (Note c)	3.8.5	4.8.5							3		*				
Fungus (Note c)	3.8.6	4.8.6								1					

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspections (Continued)

Inspection	Reqt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
Subgroup 5															
Line-to-Ground Voltage	3.7.1.2	4.7.1.2**										1			
Clock/Data Phasing	3.7.1.3	4.7.1.3										2			
Offset	3.7.1.4	4.7.1.4**										3			
Slew Rate	3.7.1.5	4.7.1.5										4			
Clock Asymmetry	3.7.1.8	4.7.1.8										5			
Multiplexing	3.7.2	4.7.2										6			
Operation from Internal Timing	3.7.3.1	4.7.3.1										7	*		
Operation from Mastergroup	3.7.3.2	4.7.3.2										8	*		
Operation from Station Clock	3.7.3.3	4.7.3.3										9	*		
Framing	3.7.4	4.7.4										10			
Jitter	3.7.9	4.7.9										11			
Dimensional Interchangeability	3.11	4.11										12			
Prime Power	3.7.5	4.7.5										13			
Input Load Impedance	3.7.1.7	4.7.1.7										14			
6. Group D Inspection															
Reliability	3.10	4.10												8 Units	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.4 Inspections covered by subsidiary documents The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirements specified in the contract or purchase order.

	Where Required
Printed Wiring	3.2.2
Semiconductor devices and integrated circuits	3.2.3
Parts, materials, and processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsiblilty for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in table III shall be inspected for conformance to the inspections specified in table III.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in table III. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table III shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

b. Only for spares that are not installed into end equipment do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table III (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspections of step a above shall be assembled into the end equipment, which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and a 100% end equipment visual/mechanical and electrical inspection in accordance with table II, in that order.

d. After completion of the above 100% inspection by the contractor discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table III. Group A electrical inspection shall be performed in the order shown in table III.

TABLE III. Group A inspections.

Inspection	Reqt Para	Insp Para	AQL	
			Major	Minor
VISUAL-MECHANICAL (Including Spare Assemblies)				
CCA, Format Generator	3.15	4.13	2.5%	4.0%
CCA, Group Logic A	3.15	4.13	2.5%	4.0%
CCA, 5 Vdc	3.15	4.13	2.5%	4.0%
CCA, Scanner	3.15	4.13	2.5%	4.0%
CCA, Group Logic B	3.15	4.13	2.5%	4.0%
CCA, Receive Rate Generator	3.15	4.13	2.5%	4.0%
CCA, Framing	3.15	4.13	2.5%	4.0%
CCA, Mastergroup Receive	3.15	4.13	2.5%	4.0%
CCA, AC Input	3.15	4.13	2.5%	4.0%
CCA, Multi Vdc	3.15	4.13	2.5%	4.0%

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE III Group A inspections (continued)

Inspection	Reqt Para	Insp Para	AQL Major	AQL Minor
CCA, Orderwire Mux/Demux	3.15	4.13	2.5%	4.0%
CCA, BITE 1	3.15	4.13	2.5%	4.0%
CCA, Group Rate Selector Assembly	3.15	4.13	2.5%	4.0%
Case and Covers	3.15	4.13	2.5%	10.0%
ELECTRICAL PERFORMANCE (Including Spare Assemblies)				
CCA, Format Generator	3.12	4.12	1.0%	*
CCA, Group Logic A	3.12	4.12	1.0%	*
CCA, Scanner Logic	3.12	4.12	1.0%	*
CCA Group Logic B	3.12	4.12	1.0%	*
CCA, Receive Rate Generator	3.12	4.12	1.0%	*
CCA, Framing	3.12	4.12	1.0%	*
CCA, Mastergroup Receiver	3.12	4.12	1.0%	*
CCA, Orderwire Mux/Demux	3.12	4.12	1.0%	*
CCA, BITE 1	3.12	4.12	1.0%	*
CCA, 5 Vdc	3.12	4.12	1.0%	*
CCA, Multi Vdc	3.12	4.12	1.0%	*
CCA, AC Input	3.12	4.12	1.0%	*
CCA, Group RATE Selector Assembly	3.12	4.12	1.0%	*
ELECTRICAL				
BITE and Alarms	3.7.6	4.7.6	1.0%	*
Operational Test	3.7.7	4.7.7	1.0%	*

*All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection. This inspection shall conform to table IV and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B AQL shall be as shown in table IV and the inspection level shall be S-4.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in table IV.

TABLE IV. Group B inspections.

Inspection	Reqt Para	Insp Para	AQL
Line-to-Line Voltage Levels and Signal Balance	3.7.1.1	4.7.1.1	4.0%
Receiver Dynamic Range	3.7.1.6	4.7.1.6	4.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in table V. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eighth month. The unit selected every fourth month shall be subjected to vibration and shock testing in that order. The unit selected every eighth month shall be subjected to EMI testing.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every twelfth month. One unit shall be subjected to altitude, dust and salt fog testing and one unit shall be subjected to fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For each subsequent Group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in table V.

TABLE V. Group C inspections.

Inspection	Reqt Para	Insp Para
Subgroup 1		
High Temperature	3.8.1.1	4.8.1.1
Low Temperature	3.8.1.2	4.8.1.2
Subgroup 2		
Vibration/Shock	3.8.7	4.8.7
EMI	3.9	4.9
Subgroup 3		
Humidity	3.8.2	4.8.2 (Note 1)
Subgroup 4		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4 (Note 1)
Salt fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE V. Group C inspections (continued)

Inspection	Reqt Para	Insp Para
Subgroup 5		
Clock/Data Phasing	3.7.1.3	4.7.1.3
Slew Rate	3.7.1.5	4.7.1.5
Clock Asymmetry	3.7.1.8	4.7.1.8
Multiplexing	3.7.2	4.7.2
Operation from Internal Timing	3.7.3.1	4.7.3.1
Operation from Master-group Timing	3.7.3.2	4.7.3.2
Operation from Station Clock	3.7.4.3	4.7.4.3
Framing	3.7.4	4.7.4
Jitter	3.7.9	4.7.9
Dimensional Interchangeability	3.11	4.11
Prime Power	3.7.5	4.7.5
Input Load Impedance	3.7.1.7	4.7.1.7

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table VI and shall be performed on units from lots which have been subjected to and have met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE VI. Group D inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Reliability	3.10	4.10

4.5.5.1 Sampling for Group D inspection. Eight units shall be selected at random from the first month's production and subjected to the test of 4.10. For subsquent Group D inspections, a minimum of eight units shall be selected at random from every fourth month's production.

4.5.5.2 Group D failure. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass Group A inspection only.

4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning the equipment shall be connected per figure 1. Test conditions of 4.10.3 shall apply except that the duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted Failure data obtained shall be submitted with the lot containing the unit in which the failure occurred.

a. Set groups 1 through 12 to position 10 (1536 kb/s).

b. Set the MUX OW CCA to MG RATE 17, STA CLK to 32 and TMG to STA CLK.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

c. Run the units for 48 hours, using BITE to indicate failure. If a failure is detected, it is to be repaired and recorded. The last 24 hours of burn-in must be failure free.

d. S, SA and FS Alarms shall be latched to detect any failures occurring during the burn-in period.

e. After completion of burn-in, perform the operational test of 4.7.11.

4.7 Electrical. (See 3.7) Unless otherwise specified all tests shall be preformed at 115 V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipment listed in table VII or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see table VIII and figure 2) and figure 2 shall be provided by the contractor subject to approval by the Government. The Government shall be responsible for supplying the Government Furnished Equipment (GFE) required for test listed in table IX. Table X identifies the TD-1237 interface for the test figures. Unless otherwise noted all resistors in test figures shall be .5 watts.

TABLE VII. Standard test equipment.

Description	Mfg./Model No. (or equivalent)
Electronic Counter	Fluke 1912A
Volt-Ohm Meter	Simpson 260
Wattmeter	Westinghouse P 15
Attenuator	H.P. 4437A
Attenuator	Kay 3-30
Oscilloscope	Tektronix 7904
Digital Multimeter	Fluke 8050A
Frequency Synthesizer	DANA 7010
AC Power Source	Elgar 401-B501

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

TABLE VIII. Special test equipment.

Equipment Name	Drawing No.	Functional Description
Group Simulator	A3051667	The Group Simulator simulates the group data, orderwire and telemetry signals. It generates either a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any conbination. A TTL error output signal is provided for count-errors externally.
Timing Source	A3051666	The Timing Source provides four independent group outputs, switch-selectable to positions between 72 and 4608 kb/s. Also provided are four independent Station Clock/Telemetry Clock outputs, switch-selectable to 2, 16, or 32 kHz.
Timing Source (Alternate)		An alternate Timing Source is shown in figure 2 that may be used in lieu of the above.
Error Inserter	A3051665	The Error Inserter provides a means of injecting a .001 to .1% continous BER with 20% random bursts.
DVOW/DOW Simulator	A3051669	The DVOW/DOW Simulator provides a pseudo-random data generator and correlator at 2, 16, and 32 kb/s with a balanced signal interface. A TTL error output signal is provided for counting errors externally.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

TABLE IX. Government Furnished Equipment. (GFE)

Equipment	Description
Multiplexer	TD-1237(P)/G

Table X. TD-1237 interface connections.

Connector	Function	PIN LO	PIN HI	PIN Gnd	Signal Type
Group 1 thru 12	DATA IN	A	B	M	NRZ
	TIM IN	C	D	N	SQ WV
	DATA OUT	E	F	P	NRZ
	TIM OUT	G	H	R	SQ WV
Master Group	DATA IN	A	B	M	NRZ
	TIM IN	C	D	N	SQ WV
	DATA OUT	E	F	P	NRZ
	TIM OUT	G	H	R	SQ WV
Station Clock and Data Orderwire	STA CLK IN	N	P	R	SQ WV
	DOW #1 DATA IN	A	B		NRZ
	DOW #1 DATA OUT	C	D		NRZ
	DOW #1 TIM OUT	E	F		SQ WV
	DOW #2 DATA IN	G	H		NRZ
	DOW #2 DATA OUT	J	K		NRZ
	DOW #2 TIM OUT	L	M		SQ WV
	ALARM S	S	T		ALM = CLOSE CONTACT
	ALARM SA	U	V		ALM = CLOSE CONTACT
	ALARM FS	W	X		ALM = CLOSE CONTACT
Power	LOW (-) HI (+)	C	A	B	

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.1 Balanced equipment-side interface signal characteristics. (See 3.7.1)

4.7.1.1 Line-to-line voltage levels and signal balance. (See 3.7.1.1)

4.7.1.1.1 Line-to-line voltage levels. (See 3.7.1.1)

a. Connect the equipment as shown in figure 3. Set each UUT Group Rate Select switch to its maximum rate.

GROUP LOGIC A- 4608 kb/s RATE 14

b. Set the MUX OW CCA MG RATE to Position 17, STA CLK to 32 and TMG to STA CLK.

c. For each of the following signals, measure the positive and negative voltage levels line-to-line (i.e., between the high and low pins):

GROUP 1 through 12 DATA OUT
MG DATA OUT
Group 1 through 12 TIMING OUT
MG TIMING OUT
DOW 1 DATA OUT
DOW 2 DATA OUT
DOW 1 TIMING OUT
DOW 2 TIMING OUT

NOTE: Measure the Master Group (MG) data first, then connect the loopback connector. When measuring groups 1 through 12, set the Group Simulator to the corresponding unit GROUP RATE. Polarity is measured on the pin with the later letter of the alphabet. The pin with the earlier letter is used as reference.

d. Verify that the following requirements are met for each signal: $\pm$ 100 mV min. and $\pm$3V max.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.1.1.2 Signal balance. (See 3.7.1.1)

a. Connect the equipment as shown in figure 3. Set the UUT GROUP RATE SELECT switches to the maximum rate .

GROUP LOGIC A- 4608 kb/s RATE 14

b. Set the MUX OW CCA MG RATE to 17, STA CLK to 32 and TMG to STA CLK.

c. For each of the signals listed in step c of 4.7.1.1.1, measure the positive and negative voltage levels line-to-line (i.e., between the high and low pins).

NOTE: Measure the MG data first, then connect the loop back connector. When measuring Groups 1 through 12, set the Group Simulator to the corresponding GROUP RATE. Polarity is measured on the pin with the later letter of the alphabet. The pin with the earlier letter is used as a reference.

d. Compare the positive and negative values. Determine the larger of the two values.

e. Compare the difference between the positive and negative values to 10% of the larger value.

f. Verify that the difference is less than 10% of the larger value.

4.7.1.2 Line-to-ground voltage. (See 3.7.1.2) Measure the line-to-ground voltage for each line using test configuration in figure 3. Verify that the requirements of 3.7.1.2 are met.

4.7.1.3 Clock/data phasing. (See 3.7.1.3)

a. Connect the equipment as shown in figure 4. The UUT and GROUP Simulator settings are the same as in 4.7.1.1.

b. For each of the signals listed in step c of 4.7.1.1.1, measure the delay between the zero crossing points on both the data rise and fall time slopes to the zero crossing points of the timing rise and fall slopes as indicated in figure 5.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

NOTES: Measure the MG signals first, then connect the loopback connector.

c. Verify that the specified requirements of 3.7.1.3 are met.

4.7.1.4 Offset. (See 3.7.1.4)

a. Connect the equipment as shown in figure 6.

b. Set each GROUP RATE SELECT switch to its maximum rate.

GROUP LOGIC A 4608 kb/s RATE 14

c. Set the MUX OW CCA MG RATE to 17 (HIGH), the STA CLK to 32 and the TMG to STA CLK.

d. Measure the dc offset voltage between high and low pins of the Group Data and Timing Out signals.

e. Verify that the DC offset voltage are not greater than ±3 volts.

4.7.1.5 Slew rate. (See 3.7.1.5)

a. Connect the equipment as shown in figure 7. The UUT and GROUP Simulator settings are the same as in 4.7.1.1.1.

b. For each of the following signals, measure the time between the +40 mV and -40 mV points around the zero voltage level of the rise and fall slopes as shown in figure 8 the following:

GROUP 1 through 12 DATA OUT
GROUP 1 through 12 TIMING OUT
DOW 1 DATA OUT
DOW 2 DATA OUT
DOW 1 TIMING OUT
DOW 2 TIMING OUT
MG TIMING OUT
MG DATA OUT

NOTE: All measurement shall be made through 100 feet of RG-108 cable.

c. Verify that the slew rates are within the required limits of 3.7.1.5.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.1.6 Receiver dynamic range. (See 3.7.1.6)

a. Connect the equipment as shown in figure 9. Variable attenuators will be connected in the normal signal paths during the course of the test. Adjust the attenuators to obtain a ±50 mV peak signal in each path.

b. The UUT and Group Simulator settings are the same as in 4.7.1.1.1.

c. For each of the Group Data and Timing signals 1 through 12 and the DOW Data and Timing signals 1 and 2, perform the following:

d. Connect the attenuator in each of the signal paths and adjust the attenuator to give a ±50 mV peak signal referenced to ground.

e. Observe the Error Detectors for a period of 10 seconds on the Groups, 30 seconds on the DOW and verify that there are no errors and UUT alarms.

4.7.1.7 Input load impedance. (See 3.7.1.7)

a. Connect the equipment as shown in figure 10.

b. Connect the test setup in turn to the following signal inputs:

GROUP 1 through 12 DATA IN
MG DATA IN
GROUP 1 through 12 TIMING IN
MG TIMING IN
STATION CLOCK IN
DOW 1 DATA IN
DOW 2 DATA IN

c. At each signal location, adjust the oscillator at a frequency of 1.0 kHz for an output voltage of 10.0 volts, as measured by the V1 voltmeter. Then, measure and record the V2 voltage.

d. Calculate the impedance using the formula:

$$Z\ (in) = \frac{V2}{V1} \times 100 \quad \text{ohms}$$

e. Verify that the requirements of 3.7.1.7 are met.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.7.1.8 Clock asymmetry. (See 3.7.1.8)

a. The equipment test setup is the same as in figure 4.

b. For Master Group, Group and DOW Timing signals, measure the time (A) between the rise and fall slopes at the zero voltage points. Then measure the time (B) between two successive rise slopes at the zero voltage points as indicated in figure 11.

NOTE: Measure the MG signals first, then connect the loopback connector.

c. Verify that the requirements of 3.7.1.8 are met.

4.7.2 Multiplexing. (See 3.7.2)

4.7.2.1 Single group input. (See 3.7.2)

a. Connect the equipment as shown in figure 12.

b. Set the MUX OW CCA MG RATE to 17 (HIGH), STA CLK to 32 kHz and TMG to STA CLK. The remaining control settings for the UUT are shown in table XI.

c. Apply traffic to Group 1 at each of the rates listed in table XI. At each rate, monitor the Group Error Detector for a 10 second period. Verify that there are no errors.

d. Set the MUX OW CCA to STA CLK and the STA CLK to 16 kHz.

e. Apply traffic to Group 1 at its maximum rate, verify that there are no errors for a 10 second period.

f. Set the MUX OW CCA MG RATE 16 (LOW). The remaining control settings are shown in table XI.

g. Apply traffic to Group 1 at each of the rates listed in table XI up to and including 2304 kb/s.

h. At each rate, monitor the Group Error Detector for a 10 second period. Verify that there are no errors.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE XI. Control settings for single group input tests.

CONTROL	SETTING	
GROUP RATE SELECTOR	SET GROUP 1 THUMBWHEELS AS FOLLOWS:	
GROUP LOGIC A CCA RATES	RATE in kb/s	SWITCH POSITION
	72	1
	128	2
	144	3
	256	4
	288	5
	512	6
	576	7
	1024	8
	1152	9
	1536	10
	2048	11
	2304	12
	4096	13
	4608	14
GROUP 2 THROUGH 12	Set Groups 2 through 12 thumbwheels to zero.	
STATION CLOCK	32 kHz	

4.7.2.2 Rate combinations. (See 3.7.2)

a. Connect the equipment as shown in figure 13. Four (4) group inputs will be exercised at a time. Group 1 through 4 will be exercised first, then group 5 through 8 and then groups 9 through 12.

b. Set the MUX OW CCA MG RATE to 17 (HIGH) and TMG to STA CLK.

c. Set the STA CLK to 32 kHz.

d. At each of the rate combinations listed in table XII, monitor the four (4) Group Input Error Detectors over a 10 second period. Verify that there are no errors.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE XII. Rate combinations for high Master Group rate.

Group	A	B	C	D	E
1	* 4915.2	* 4915.2	4096	* 4915.2	1536.0
2	4608	* 4915.2	* 4915.2	* 4915.2	1536.0
3	2304	512	288	1152	1536.0
4	2048	256	512	1152	1536.0
5	1152	576	576	576	1536.0
6	1024	576	576	576	1536.0
7	576	576	1536	576	1536.0
8	576	576	1152	576	1536.0
9	512	1152	2048	256	1536.0
10	288	1152	2304	1024	1536.0
11	256	2048	144	1024	1536.0
12	144	1024	256	1536	1536.0
	18,403.2	18,278.4	18,403.2	18,278.4	18,432.0

*If UUT not equipped for this rate, substitue 4608 kb/s.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

e. Set the MUX OW CCA MG RATE to 16 (LOW).

f. At each of the rate combinations listed in table XIII, monitor each of the four (4) Group Error Detectors over a 10 second period. Verify that there are no errors.

4.7.2.3 Data orderwire tests. (See 3.7.2)

a. Connect the equipment as shown in figure 14. Set the MUX OW CCA MG RATE to 16 (LOW), the STA CLK to 32 and TMG to STA CLK.

b. Set Group Rate Select switches 1 through 12 to 0 and the STATION CLOCK to 32 kHz.

c. Connect the DVOW/DOW Simulator at a 16 kb/s Traffic Rate to DOW 1. Monitor the DOW Error Detector over a 30 second period. Verify that there are no errors.

d. Repeat Step c for DOW 2.

e. Repeat this procedure, steps a through d with the MUX OW CCA MG RATE set to 17 (HIGH).

4.7.2.4 Rate justification tests. (See 3.7.2)

a. Connect the equipment as shown in figure 15. Set the MUX OW CCA MG RATE to 16 (LOW), STA CLK to 32 and TMG to STA CLK.

b. Set GROUP RATE SELECT switches 2 through 12 to 0 and Group 1 to position 12 (2304 kb/s).

c. At each of the following combinations of Station Clock and Group 1 frequency offsets, monitor the Group Simulator over a 10 second period. Verify that there are no errors.

Station Clock	Group 1
+5 ppm	-30 ppm
-5 ppm	+30 ppm

d. Set the MUX OW CCA MG RATE to 17 (HIGH) and the Group 1 RATE SELECT switch to position 14 (4608 kb/s)

e. Repeat Step c.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currenversion before use.

TABLE XIII. Rate combinations for low Master Group rate.

Group	A	B	C	D	E
1	2304	2304	2304	1536	1152
2	2304	1536	2304	2304	1152
3	1152	256	256	1152	1152
4	1024	128	288	576	1152
5	576	288	288	576	576
6	512	288	288	576	576
7	288	576	512	288	576
8	288	576	576	288	576
9	288	576	576	128	576
10	256	1152	1152	256	576
11	144	1024	72	512	576
12	72	512	144	1024	576
	9208	9216	9208	9216	9216

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.3 Timing modes. (See 3.7.3)

4.7.3.1 Operation from internal timing. (See 3.7.3.1)

a. For this test, the UUT and RCV unit must contain all GROUP LOGIC A CCA's. Connect the equipment as shown in figure 16.

b. Set the MUX OW CCA in both units to MG RATE 17 (HIGH), STA CLK to 32 and TMG to STA CLK.

c. Set all GROUP RATE SELECT switches 1 through 12 to 10 (1536 kb/s).

d. Apply traffic to the UUT at 1536 kb/s.

e. Verify illumination of LOI and S Alarms and LED on MUX orderwire.

f. Verify that the error rate is zero for 10 seconds.

g. Measure the frequency of the MASTERGROUP TIMING output at the MASTERGROUP connector.

h. Verify that the TIMING is between 18,719,906 and 18,720,094.

4.7.3.2 Operation from receive mastergroup timing. (See 3.7.3.2)

a. Set the UUT MUX OW CCA TMG to LOOP.

b. Apply traffic to the UUT at 1536 kb/s. Monitor the Group Error Detector over a 10 second period. Verify that there are no errors and verify that there are no alarms.

4.7.3.3 Operation from station clock. (See 3.7.3.3) The test for operation from the Station Clock is performed as part of 4.7.3.

4.7.4 Framing. (See 3.7.4)

a. Connect the equipment as shown in figure 17.

b. Set the MUX OW CCA MG RATE to 17 (HIGH), the STA CLK to 32 and the TMG to STA CLK.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

c. Set Group 1 through 4 RATE SELECT switches to 14 (4608 kb/s) and 5 through 12 to OFF.

d. Set the STATION CLOCK to 32 kHz.

e. Set the Group Simulator to 4608 kb/s for Groups 1, 2, 3 and 4.

f. Set the Error Inserter Test Set for a 10^{-3} error rate. Ensure that the Error Inserter Test Set is operating correctly by verifying that the BER on Group 1 is approximately 0.1%.

g. Squelch Mastergroup Data at the Error Inserter Test Set to cause loss-of-frame. Release squelch to restore data. Measure time from resumption of data to frame-in. Use the squelch release signal to start time-interval measurement and the signal at I/O 42 of the FRG CCA of the UUT to stop time-interval measurement. Take 50 measurements.

NOTE: FAULT LED on FRG CCA, SUMMARY FAULT and LOI lights will come on.

h. Set the MASTERGROUP RATE SELECT switch on the UUT MUX OW CCA to 16 (LOW) and set the Group Simulator and Groups 1 through 4 to a 2304 kb/s rate.

i. Repeat Step g.

j. Verify that the requirements of 3.7.4 are met.

4.7.5 Prime power variation. (See 3.7.5)

a. Connect the equipment as shown in figure 18. All Groups must contain GROUP LOGIC A CCA's.

b. Set the UUT MUX OW CCA MG RATE to 17 (HIGH), the STA CLK to 32 and the TMG to STA CLK.

c. Set GROUP RATE SELECT switches 1 through 12 to 10 (1536 kb/s).

d. For each of the following combinations of voltage and frequency measure the Input Power and measure the Group BER.

115 V	60 Hz
103.5 Vac	47.5 Hz
	420 Hz
126.5 Vac	47.5 Hz
	420 Hz

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

e. Verify that the power is 200 watts max and there are no errors in 10 seconds, in each case.

4.7.6 Built-in test equipment (BITE) and alarms. (See 3.7.6) There are two (2) configurations of units. One contains 12 Group Logic A CCAs, the second contains 10 Group Logic A and two Group Logic B CCAs. For test purposes the units will be configured either with the 12 Group Logic A CCAs in Groups 1 through 12 or with two Group Logic B CCAs in Groups 1 and 2 and 10 Group Logic A CCAs in Groups 3 through 12.

a. Connect the equipment as shown in figure 19. All groups must contain Group Logic A CCAs.

b. Set the UUT MUX OW CCA MG rate to 17 (HIGH), the STA CLK to 32 and the TMG to STA CLK.

c. Set the Group Rate Select switches 1 through 12 to 7 (576 kb/s).

d. Turn the power ON and verify that the following BITE requirements are met.

S OPEN
SA OPEN
FS OPEN
SUMMARY FAULT lamp must be OFF
POWER OUT-OF-TOLERENCE (POT) lamp must be OFF
All CCA LEDs must be OFF

e. Disconnect the MASTERGROUP loopback plug on the UUT.

f. Verify that the following BITE requirements are met:

S CLOSED
SA OPEN
FS OPEN
SUMMARY FAULT Lamp ON, LOI Light ON

g. Reconnect the MASTERGROUP loopback plug on the UUT.

h. Place the Framing CCA on an extender and ground connector pin 23.

i. Verify that the following BITE conditions are met:

S CLOSED
SA OPEN
FS CLOSED
LOI LED OFF
FRAMING LED lights
SUMMARY FAULT lamp ON

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

j. Remove ground from CCA pin 23, and replace the Framing CCA into the UUT.

k. Press the LAMP TEST switch on the UUT.

l. Verify that the following BITE requirements are met:

S CLOSED
SA CLOSED
FS CLOSED
SUMMARY FAULT lamp must light, POT must light

m. Turn OFF the power to the UUT at the source.

n. Verify that the following BITE requirements are met.

S CLOSED
SA CLOSED
FS OPEN

o. Turn off the UUT Power Switch.

p. Verify that the following BITE requirements are met.

S CLOSED
SA OPEN
FS OPEN

4.7.7 Operational check. (See 3.7.7)

a. Connect the equipment as shown in figure 20. The UUT is to be configured with one Logic B CCA in Group 1 and Logic A CCAs in Group 2 through 12.

b. Set the UUT MUX OW MG RATE to 17 (HIGH), the STA CLK to 32 and the TIMING to STA CLK.

c. Set the Group Rate Select switch 1 to 15 (4915.2 kb/s) and switches 2 through 12 to 7 (576 kb/s).

4.7.7.1 Group Data Error Check. (See 3.7.7.1)

a. Turn the UUT POWER switch ON.

b. Monitor the Group Error Detectors for a period of 10 seconds. Verify that there are no errors.

kb/s) and switches 2 through 7 (576 kb/s).

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.7.2 DATA OW Check. (See 3.7.7.2) Monitor the Data OW Error Detector for a period of 30 seconds. Verify that there are no errors.

4.7.7.3 Alarm Monitor Check. (See 3.7.7.3) The tests in 4.7.7 satisfy this testing requirement.

4.7.8 Environmental operational test. (See 3.7.8)

a. Connect the equipment as shown in figure 21.

b. Set Groups 1 through 12 to position 10 (1536 kb/s) on the UUT and the Auxiliary Unit.

c. Set the MUX OW CCA on the UUT to MG RATE 17, STA CLK to 32 and TMG to LOOP.

d. Set the MUX OW CCA on the Auxiliary Unit to MG RATE 17, STA CLK 32 and TMG to STA CLK.

4.7.8.1. Group data error check. (See 3.7.8.1) Monitor the two Group Error Detectors over a 10 second period. Verify that there are no errors.

4.7.8.2 Data OW error check. (See 3.7.8.2) Monitor the Group 1 and 2 Data OW Error Detector for a period of 30 seconds. Verify that there are no errors.

4.7.8.3 Alarm monitor check. (See 3.7.8.3)

a. Monitor the S, SA, LOI, and FS alarms. Verify that there are no alarms generated.

b. Disconnect Group 1 cable from the Auxiliary Unit and verify that there are S and LOI alarms but no SA alarm.

4.7.8.4 Sync/frame check. (See 3.7.8.4)

a. Monitor the Summary Alarm and LOI indicators on the UUT while turning off the equipment outside the test chamber. The UUT should indicate a Summary Alarm and an LOI.

b. Monitor the Summary Alarm indicator on the UUT while energizing the equipment outside the test chamber. The Summary Alarm Lamp and LOI should extinguish indicating that the UUT has framed up.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.9 Jitter. (See 3.7.9)

a. Connect the equipment as shown in figure 22 using the dotted line connections from the synthesizer's 1 MHz reference and output ports to the jitter meter inputs. Set the jitter meter to CAL.

b. Set the synthesizer output for 1.00000125 MHz. This will provide a square wave representing a ±1.25 ppm jitter measurement; the square wave will appear on the oscilloscope display and will be stored for reference.

c. Set the Group Simulator to EXT CLK, allowing the synthesizer frequency of 4.608000 MHz to be used. The timing source must be switched to REF to enable synchronization with the Group Simulator. Set the Jitter Meter switch to TEST. Set the equipment Group 1 rate switch to 14. The jitter frequency signal will appear on the Nicolet scope. If the jitter is within the desired calibration (±1.25 ppm) square wave, then the specification is met. A desirable sweep speed for the oscilloscope is 100 microseconds.

d. Repeat Steps b and c with the synthesizer output set to 1.000002 MHz. This will provide a square wave representing ±2 ppm on the oscilloscope.

e. Set the Group Simulator to INT CLK, and the Timing Source to INT. The synthesizer is not needed after the calibration curve is taken for ±2 ppm. The Nicolet display will show the frequency jitter in a signal form which must stay within the calibration square wave.

4.8 Service conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2 Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +63°C (+145°F). Maintain internal chamber temperature for 6 hours at +63°C (+145°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +63°C (+145°F) within a time period of one hour. Repeat the above steps two additional times for a total of three 12-hours cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +63°C (+145°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.8.1.2 Low temperature. (See 3.8.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C(-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation. (Para. II- 3.2 of Method 502.2). The chamber temperature in step 1 of Procedure II shall be adjusted to -31.7°C(-25°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III.

4.8.3 Low pressure (altitude). (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (20.6 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.

4.8.4 Dust. (See 3.8.4) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. In step 2 and step 5, the air velocity shall be 200 ±100 feet per minute. In step 5, the air temperature shall be adjusted to +63°(+145°F). The equipment shall be reoriented periodically to expose all its sides to the dust stream. The equipment shall be operated before step 1, during step 8 and during step 11.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C(+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall be no clogging or binding of moving mechanical parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs, etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Vibration and shock. (See 3.8.7)

4.8.7.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3, Procedure I, Category 8 (Ground Mobile). The vibration level shall be in accordance with figures 514.3-7 through 514.3-9 of MIL-STD-810D. The axes designated on the schedules are the vehicle axes. The orientation of the equipment relative to the vehicle, therefore, must be considered for proper direction of vibration input. Test time shall be 4 hours in each of three mutually perpendicular directions for a total of 12 hours. Prior to and following these tests, the equipment shall be operated in accordance with MIL-STD-810D general test performance guidance paragraph 4.5. The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

4.8.7.2 Shock.

4.8.7.2.1 Functional. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure I. The shock pulse shall be 20g peak for a nominal duration of 11 msec in accordance with figure 516.3-4.

4.8.7.2.2 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government-approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First article reliability test. First article reliability testing shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. Two Multiplexers TD-1237/G shall be operated back-to-back.

b. Duty cycle. Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24-hour cycle.

c. Chamber temperature shall be 40±5°C (95°F to 113°F).

d. Vibration shall be 2.2 g ±10% peak acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.10.4 Reliability test method. The detailed test procedures, and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor-prepared, Government-approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.8.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781C. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings listed in 3.12.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.13 Visual and mechanical. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (System safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System safety engineering.)

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection . All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table XIV, "Packing Inspection Provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and AQL of 4.0% defective.

TABLE XIV. Packing inspection provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

4.15 Serial numbers. The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and CCA.

4.16 Nuclear tests. Tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, tables I through VII, coded as follows:

5.1.1 Level A. 3Q-1-00-00-NR-X-ED.

5.1.2 Level B. 2M-1-00-00-NR-X-ED.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The equipment will be employed in the AN/TSQ-111 and in the radio park. It will be used in conjunction with the Multiplexer, TD-1235(P)/TTC, the Modem, Cable Driver, MD-1025/G, the Multiplexer, TD-1236/G, the Modem, Digital Data, MD-1026(P)/G, the Modem, Cable Driver, MD-1023(P)/G, the Modem, Cable Driver, MD-1024/G and the Trunk Encryption Device (TED), KG-81. The unit is intended for use in shelters and vans, and it is to be mounted in an equipment case.

6.1.1 Functional description. The equipment will capable of combining up to 12 group or supergroup inputs into a mastergroup output and decombining a mastergroup input into up to 12 group or supergroup outputs. It will accept group and supergroup bit rates ranging from 72 to 18,720 kb/s. The equipment will operate asynchronously and be capable of accepting inputs which have not been slaved into a single timing source. It will provide two orderwire channels as part of the overhead. Each will be capable of operating at 16 kb/s.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level of preservation and packing (See Section 5).

c. When first article rough handling tests are not required.

d. Quantity of first article units required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirement.

i. MIL-Q-9858.

j. Length of cable (if any are required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. (See 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. (See 4.5.3.3) Initial shipments of end items will be contingent upon successful completion of all first article tests.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

6.5 Location of operational inspection. (See 4.7.8) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table IX.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject terms (key-words) list.

Asynchonous
Balanced Equipment-side interface
Digital Group Multiplex
Framing
Group
Jitter
Master Group
Master Group Multiplexer
Multiplexer
Multiplexer, TD-1237/(P)/G
Multiplexing
Orderwire
Telemetry

CUSTODIAN:
ARMY-CR

Preparing Activity
ARMY-CR
Project Nr. 5820-A833

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

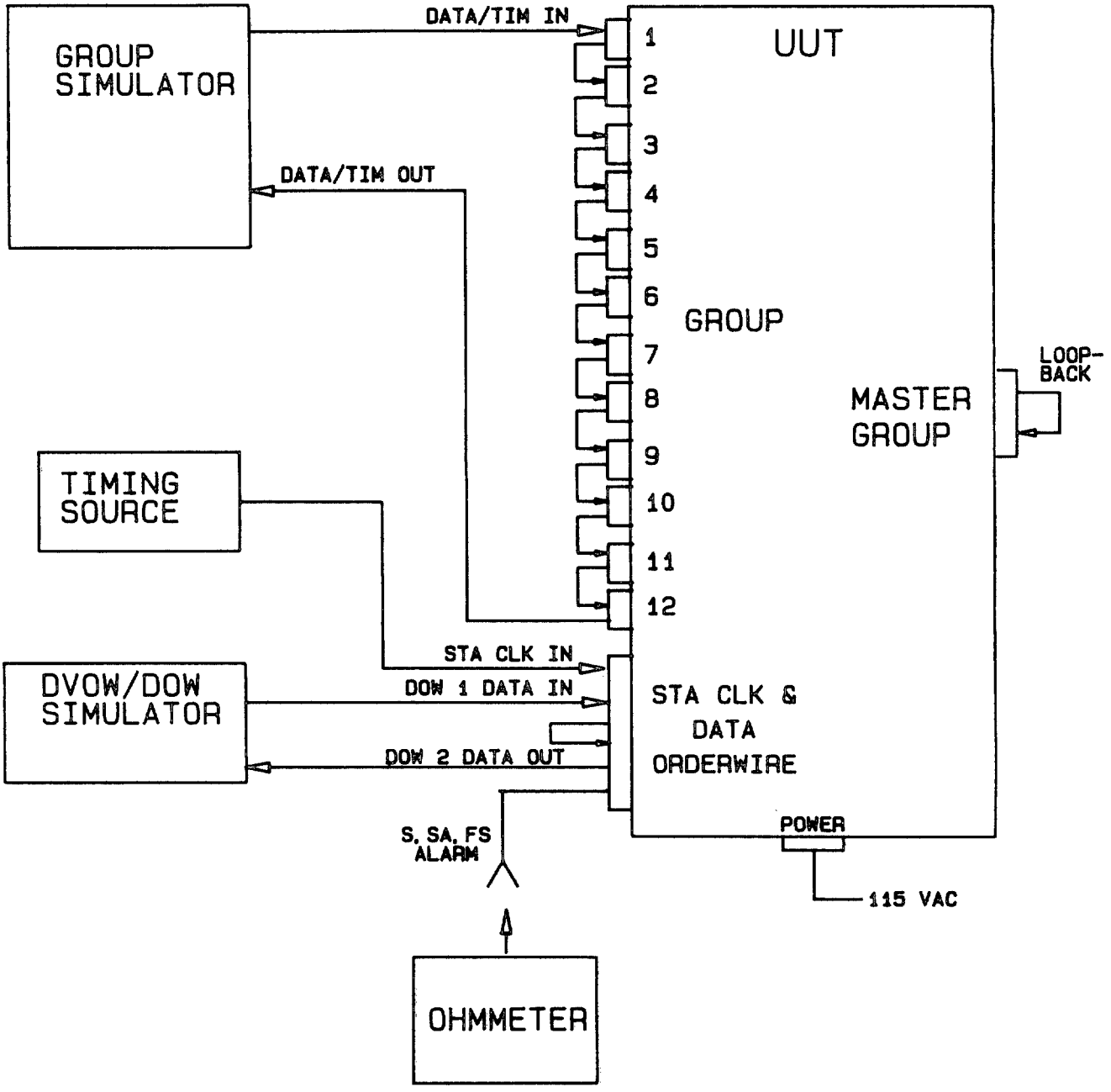
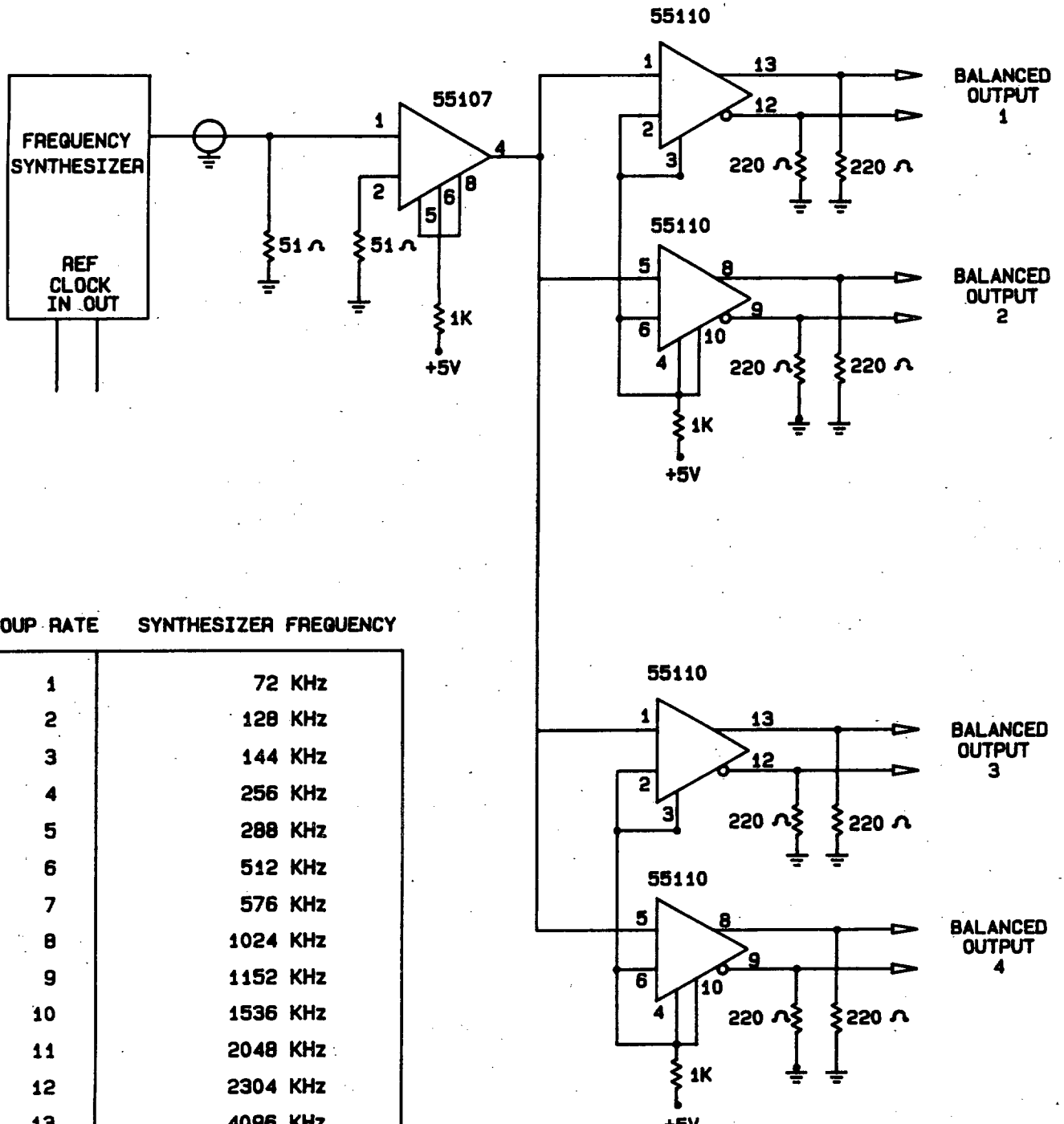


FIGURE 1. Burn-in.

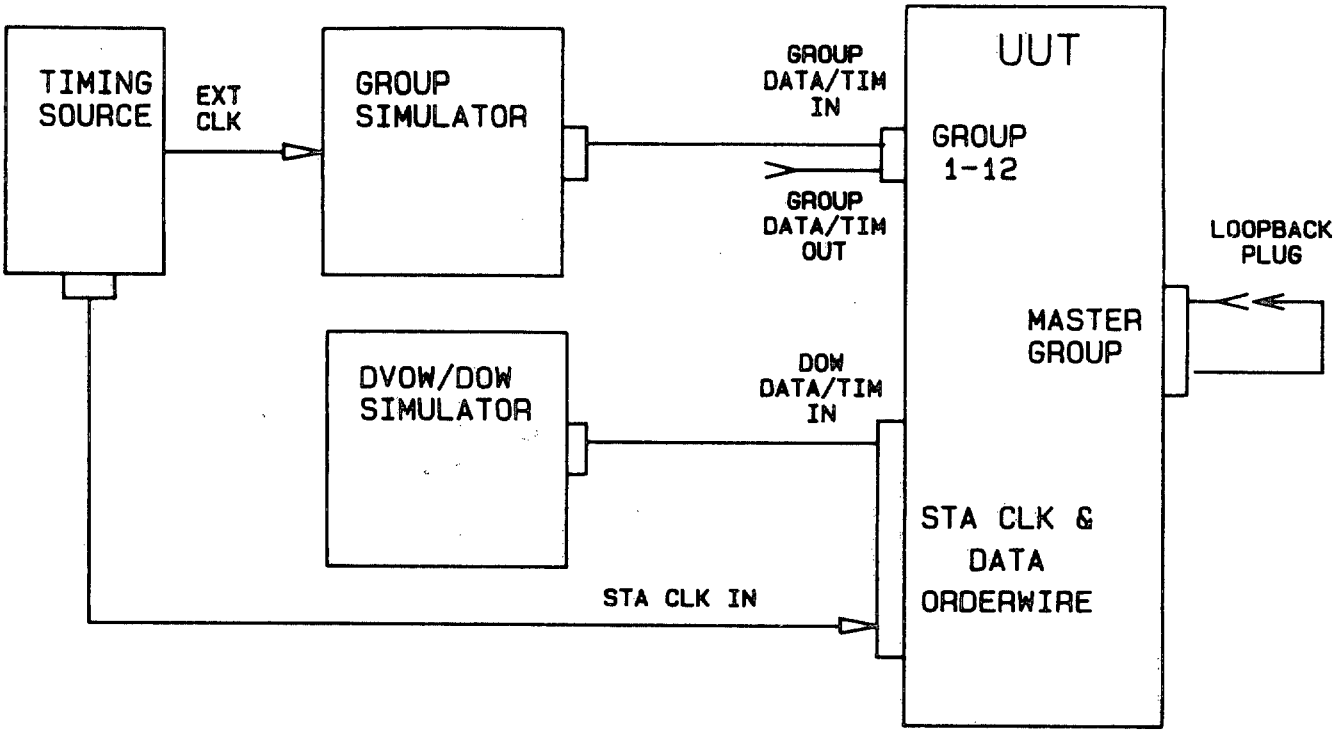
Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

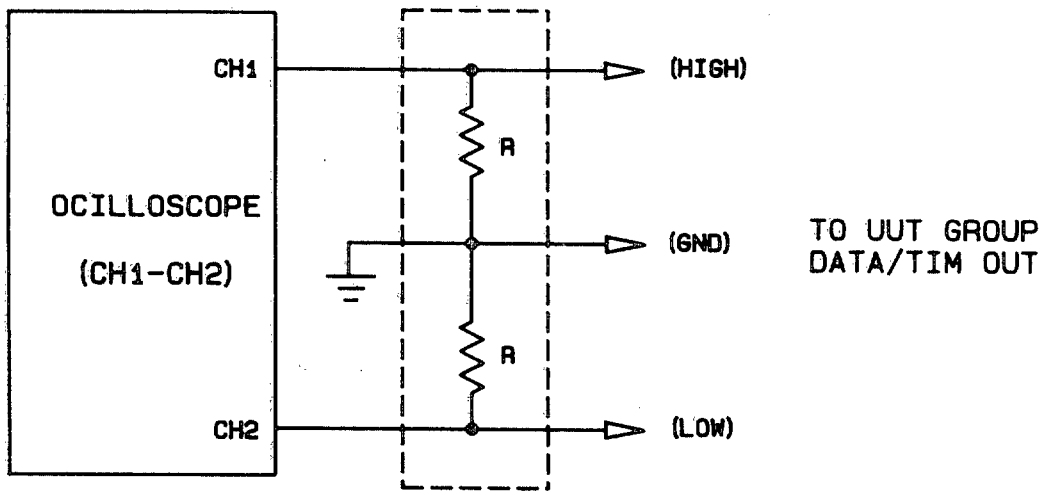


GROUP RATE	SYNTHESIZER FREQUENCY
1	72 KHz
2	128 KHz
3	144 KHz
4	256 KHz
5	288 KHz
6	512 KHz
7	576 KHz
8	1024 KHz
9	1152 KHz
10	1536 KHz
11	2048 KHz
12	2304 KHz
13	4096 KHz
14	4608 KHz
15	4915.2 KHz

FIGURE 2. Alternate timing source.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

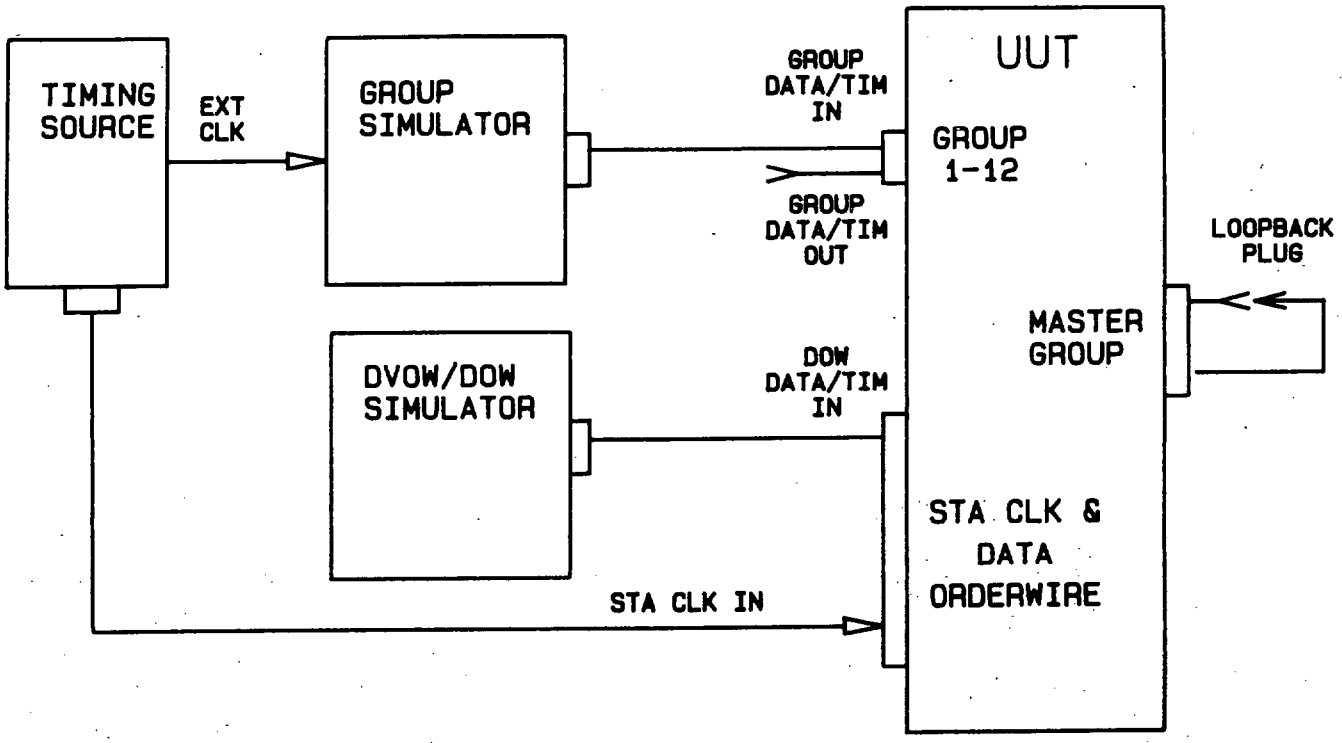


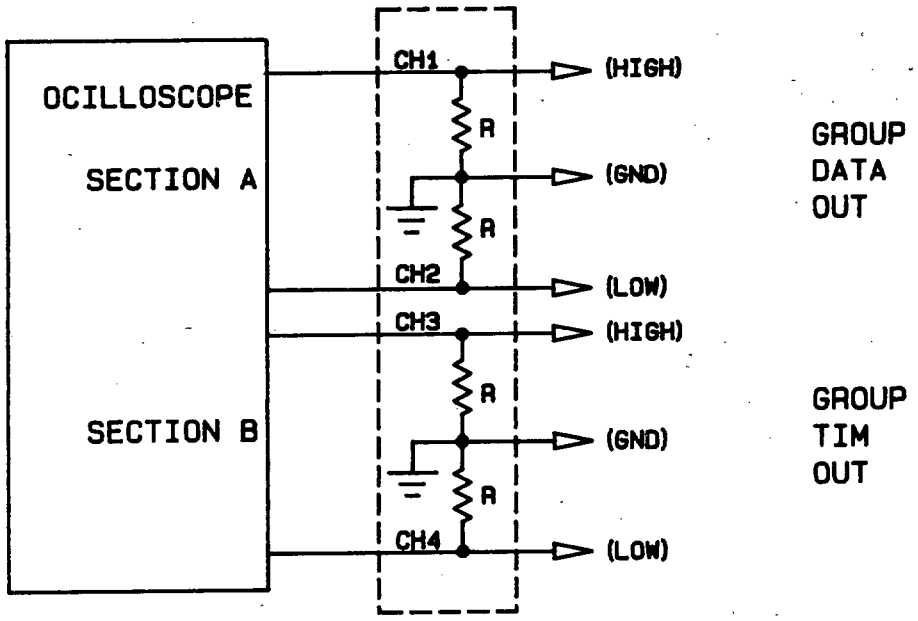


R 50 OHMS ± 1%

FIGURE 3. Line-to-line voltage levels and signal balance.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.





R 50 OHMS ± 1%

FIGURE 4. Clock/data phasing.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

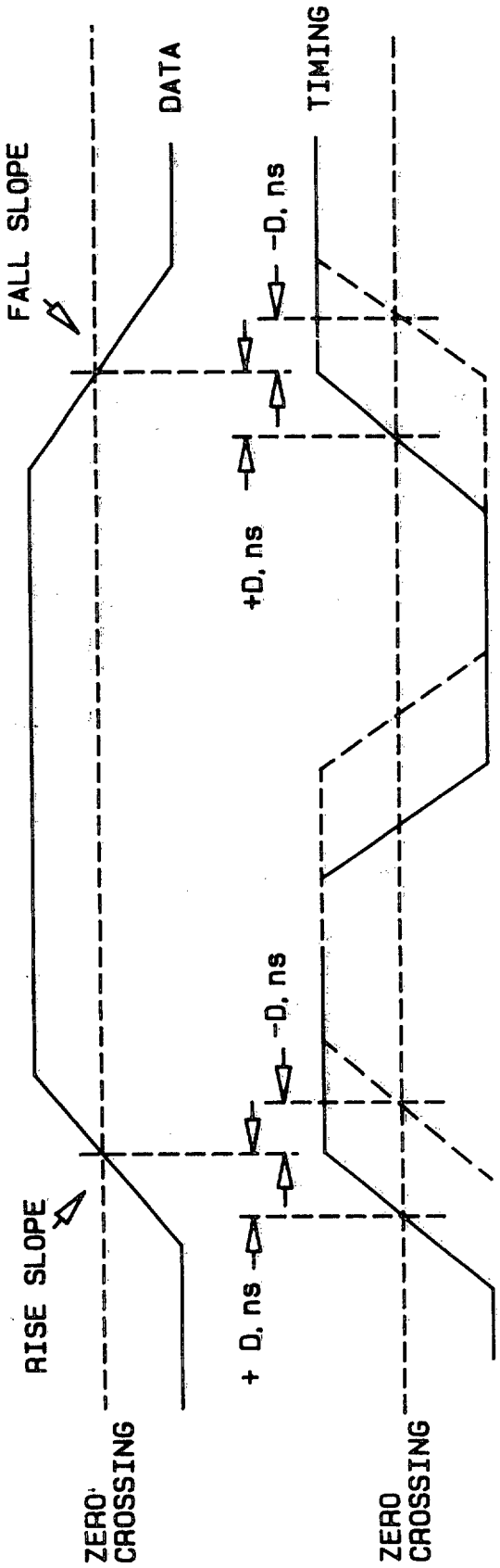
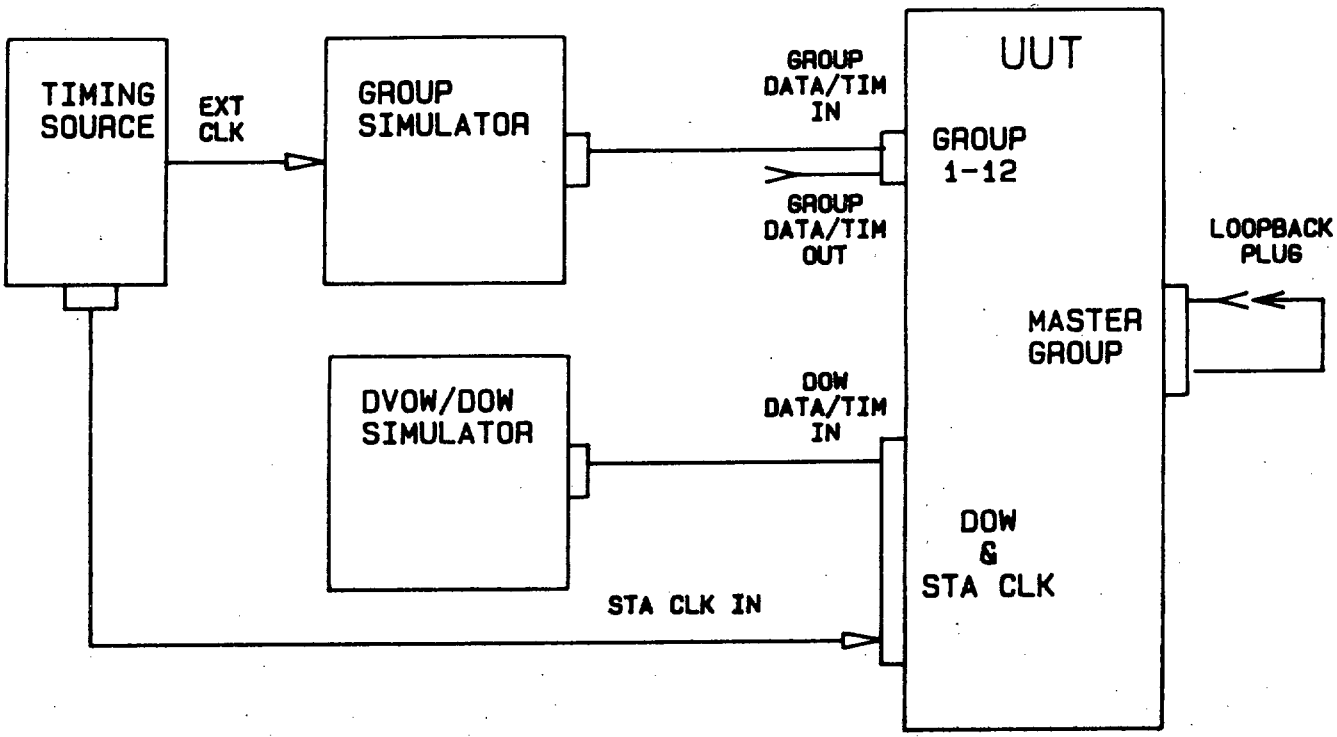


FIGURE 5 . Delay measurement (clock/data phasing).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.



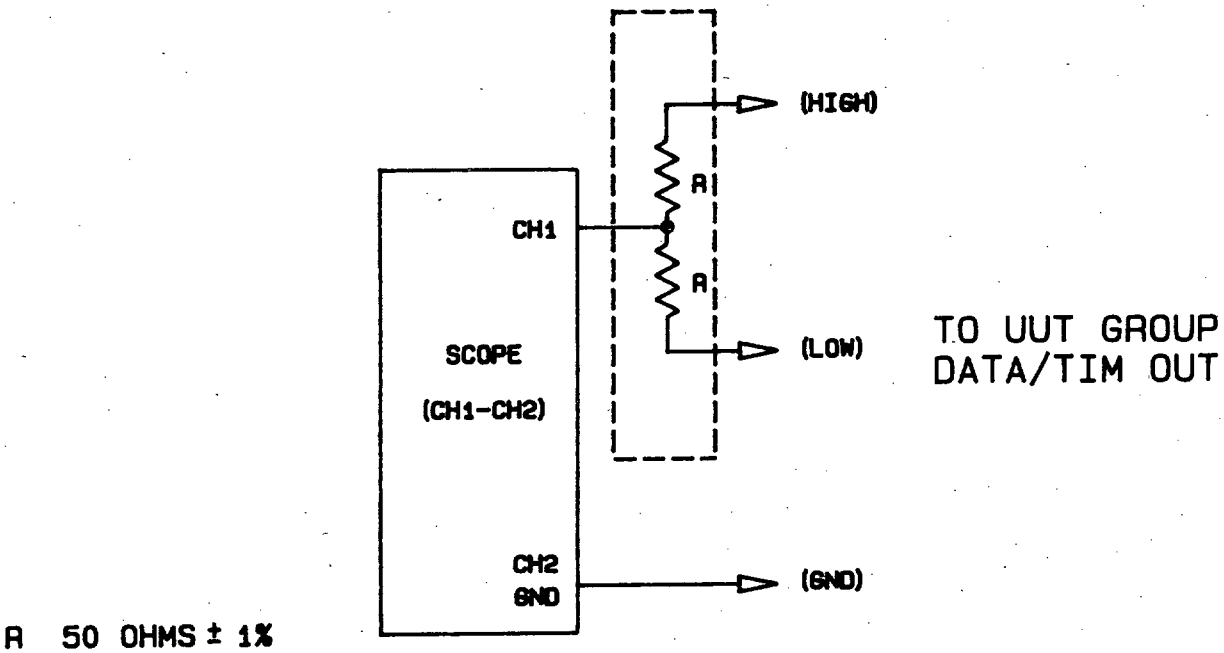
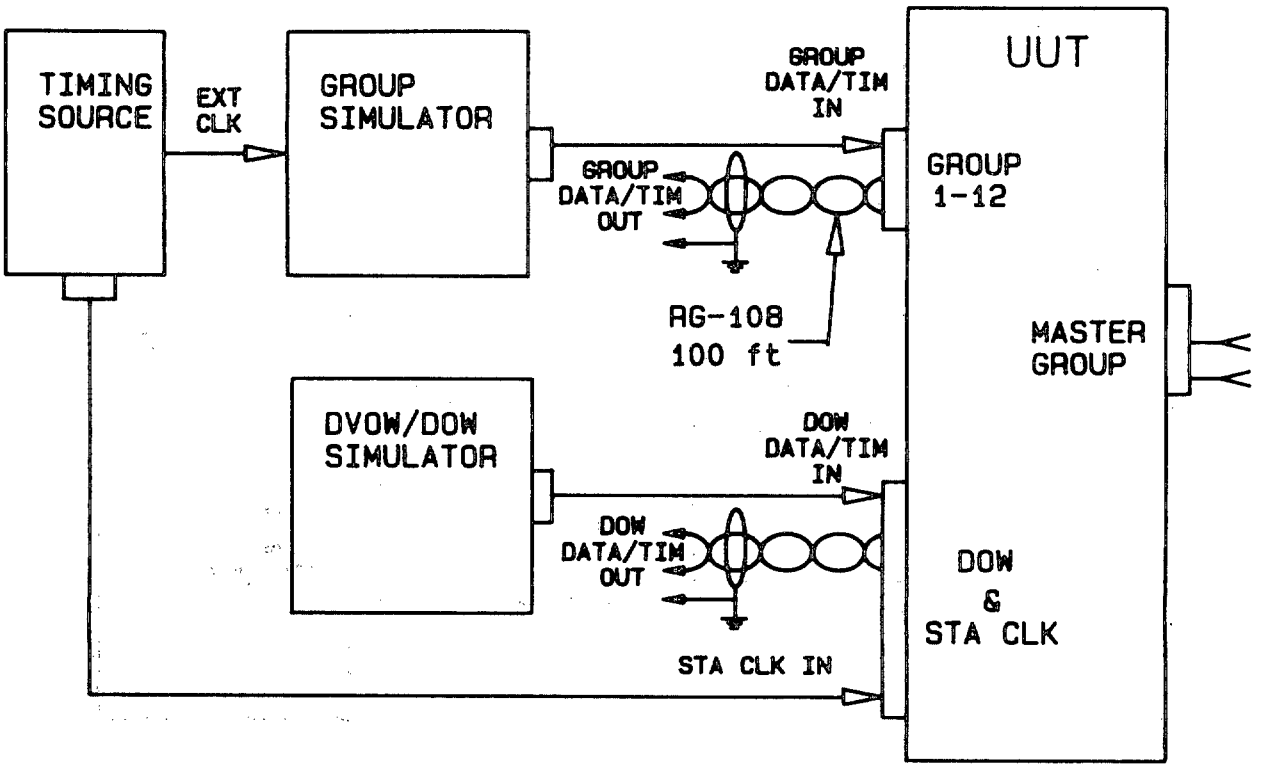
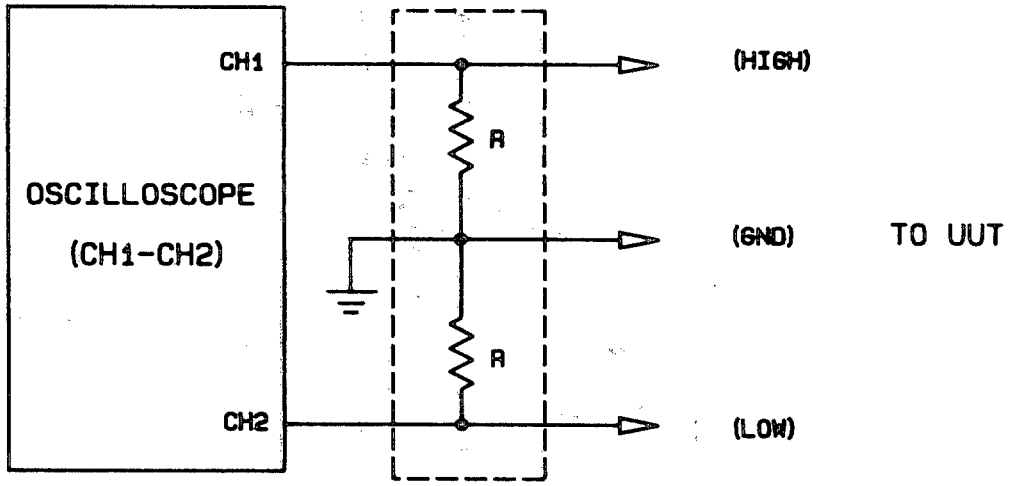


FIGURE 6. Offset voltage.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.





R 39 OHMS ± 1%

FIGURE 7. Slew rate.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

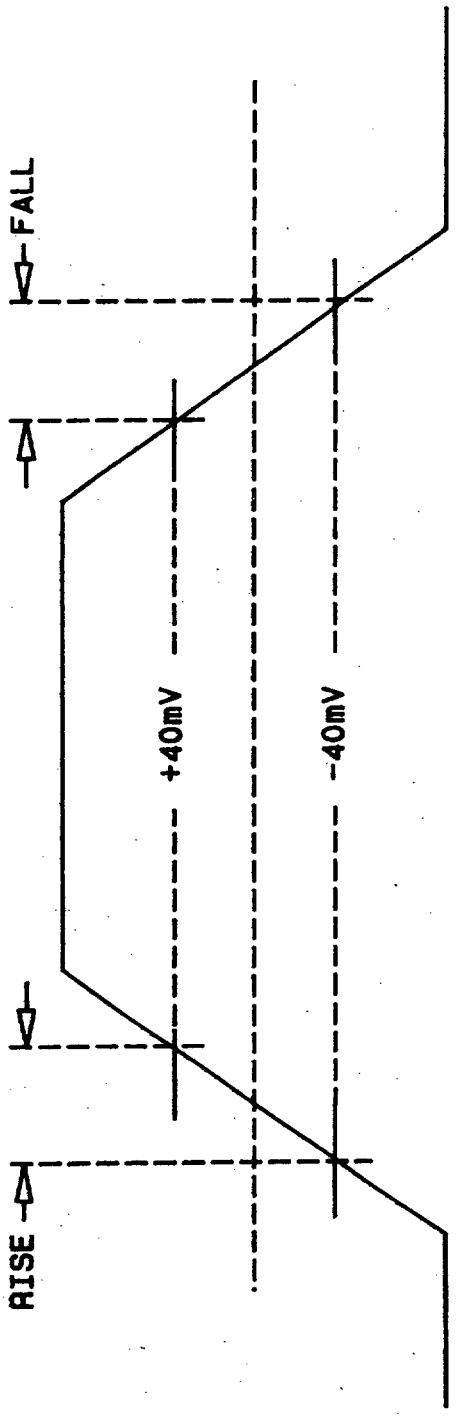


FIGURE 8 . Rise and fall slopes (slew rate).

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

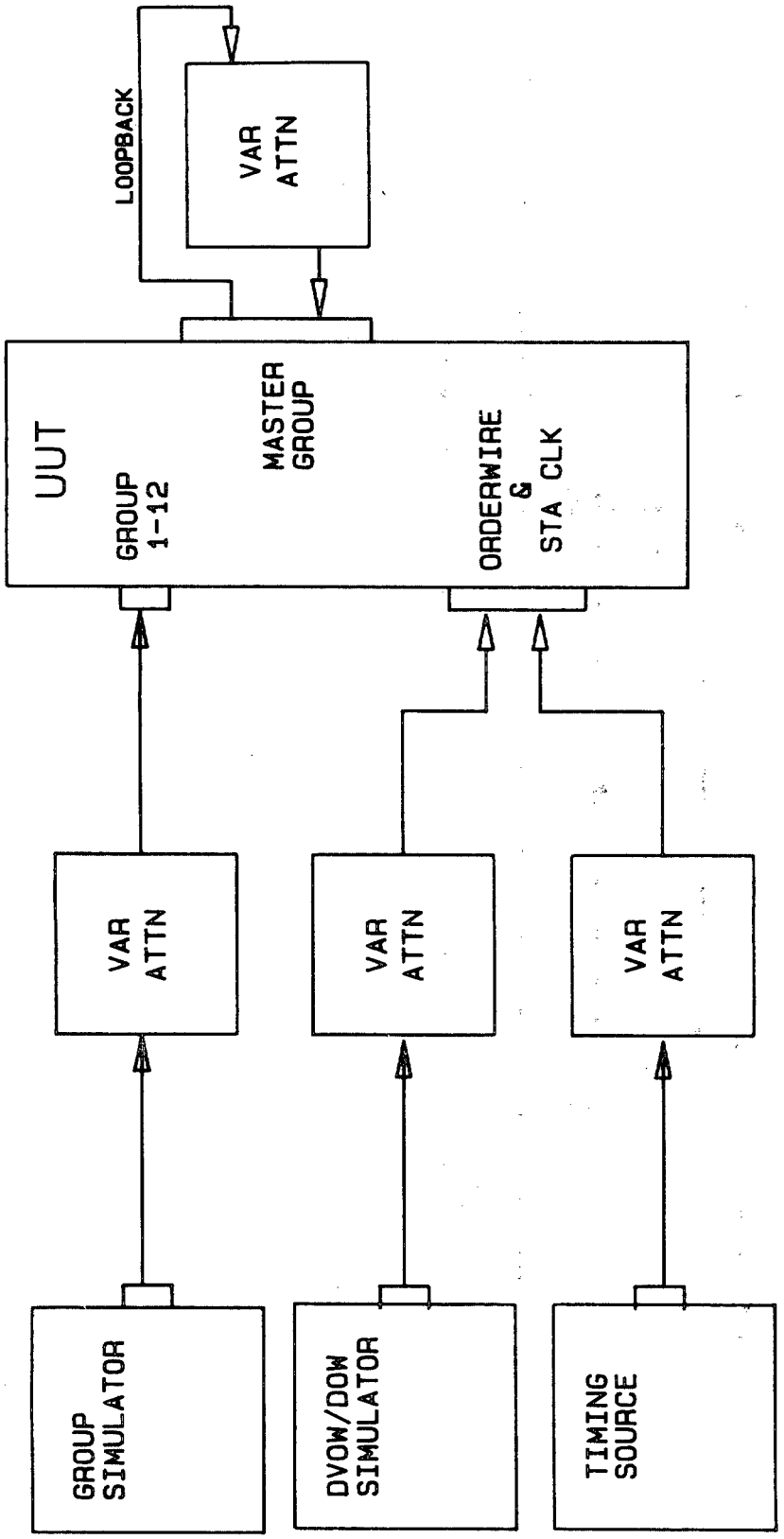
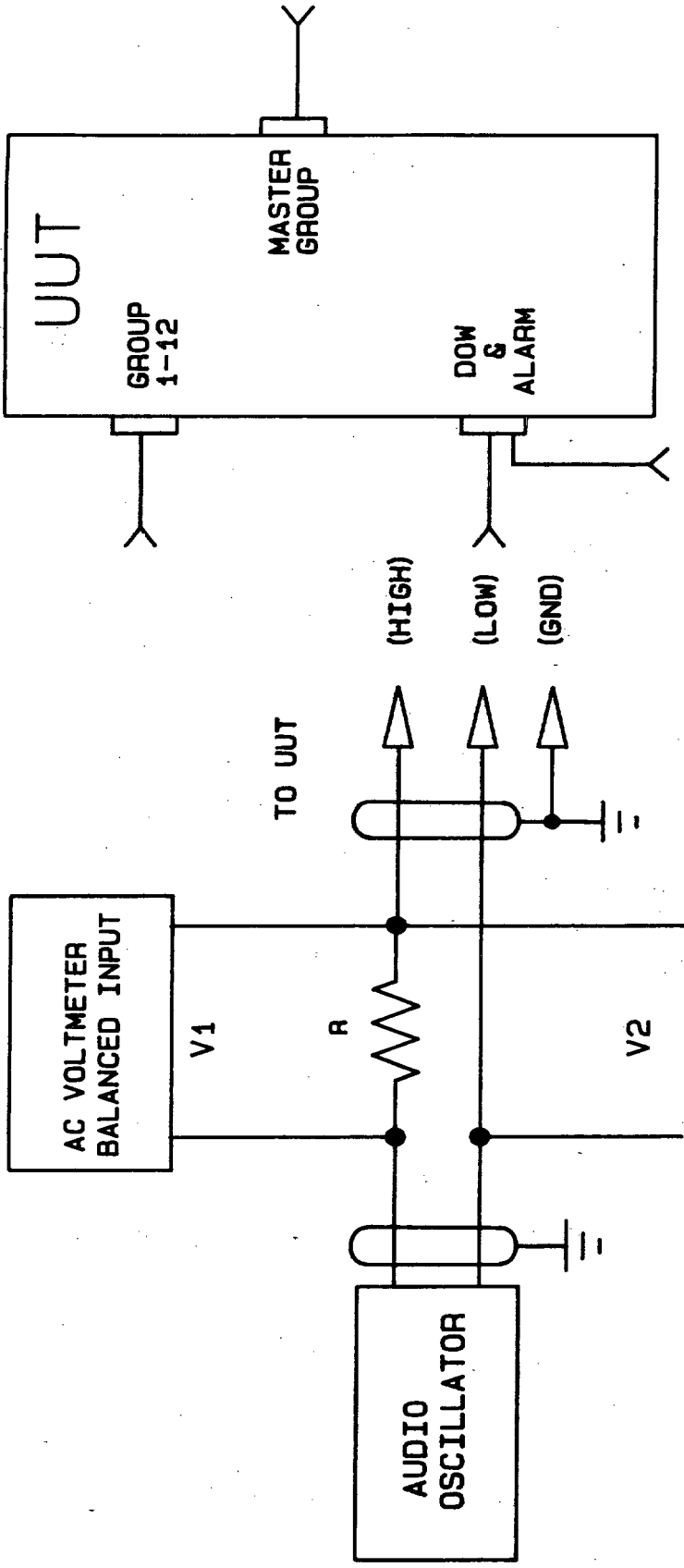


FIGURE 9. Receiver sensitivity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.



R 100 OHMS ± 1 %

FIGURE 10. Input load impedance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

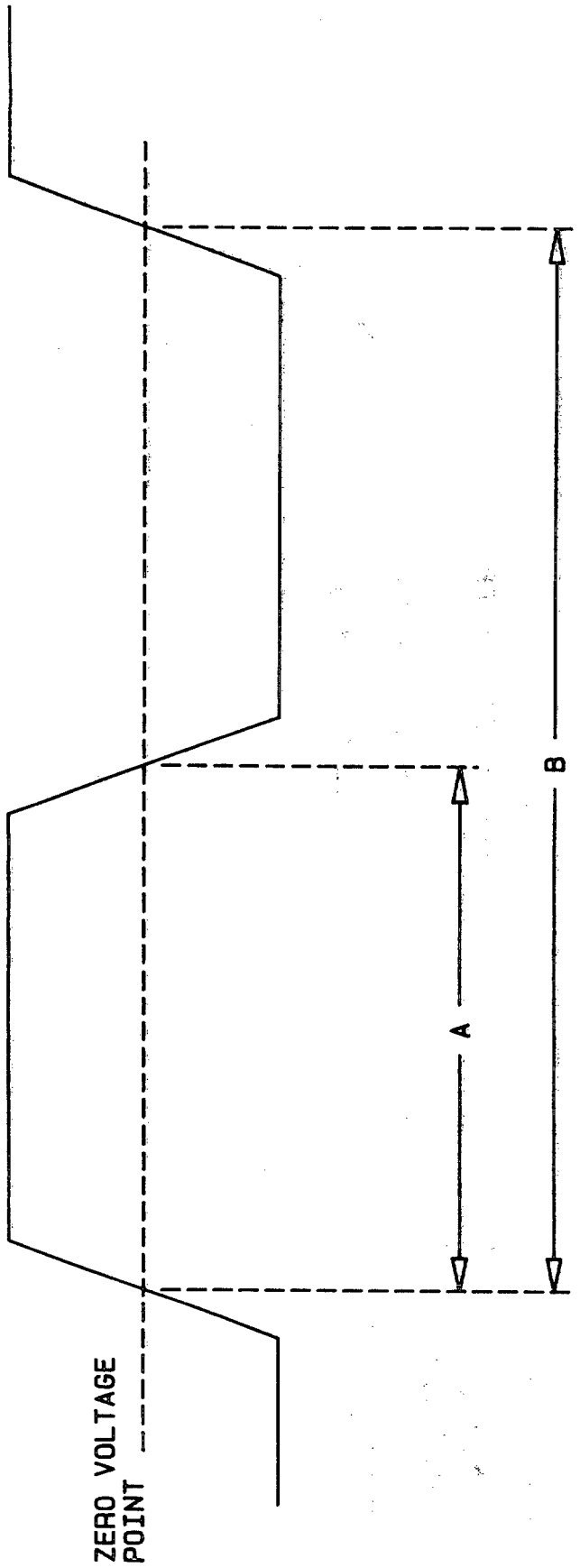


FIGURE 11. Clock asymmetry.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

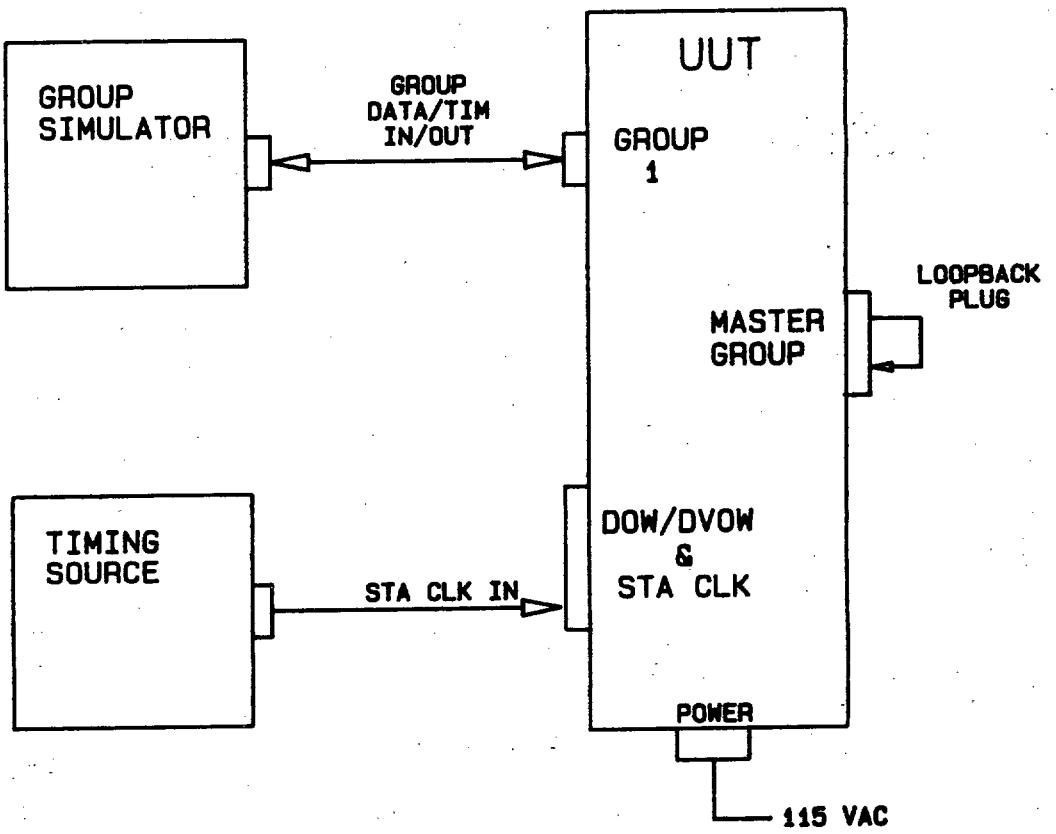


FIGURE 12. Single-group inputs.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

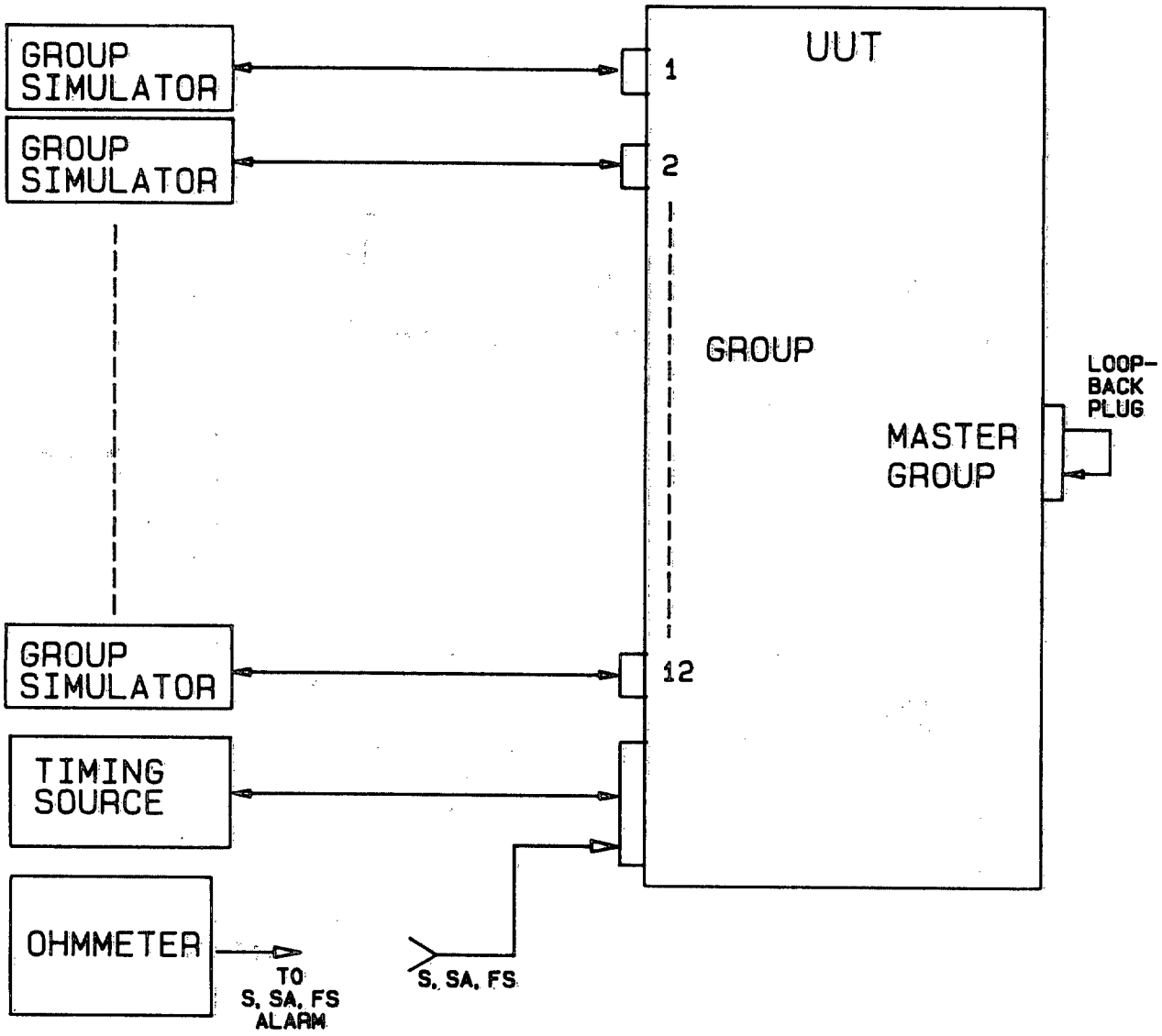


FIGURE 13. Multiple-group inputs.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

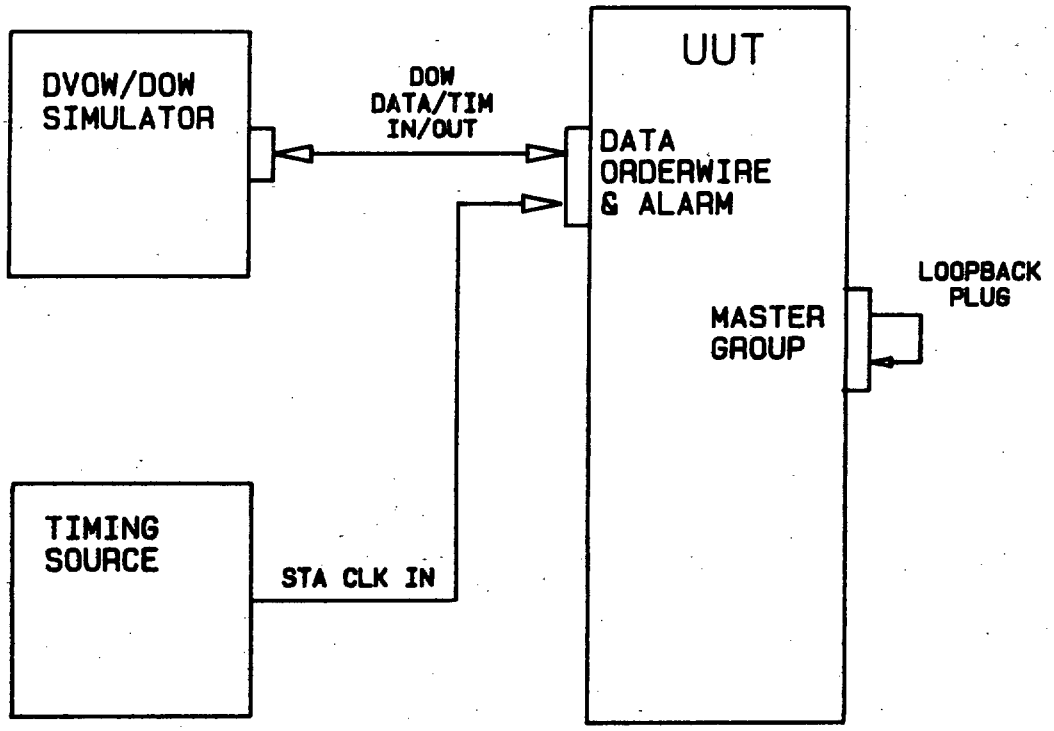


FIGURE 14. Data orderwire.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

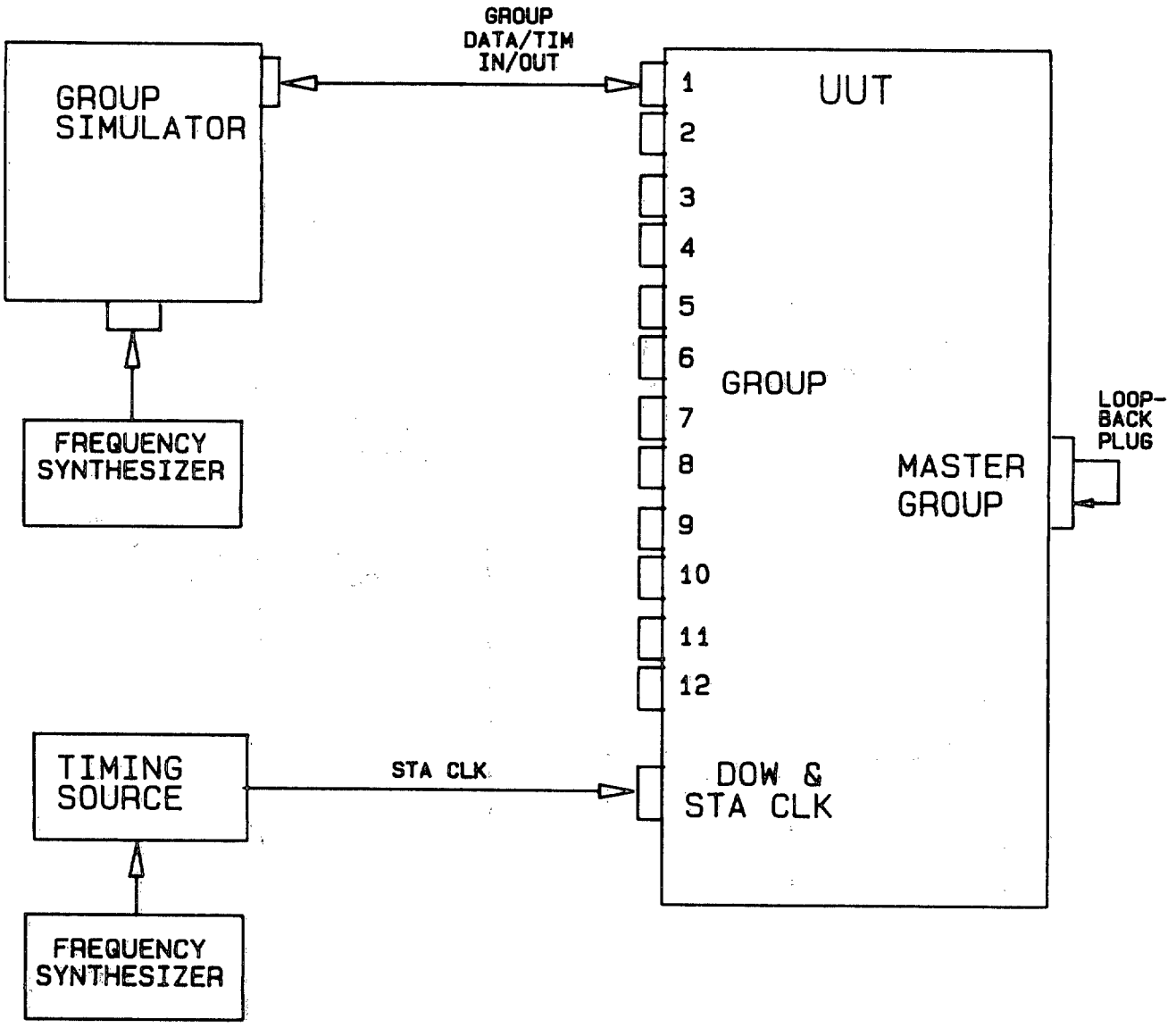


FIGURE 15. Rate justification.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

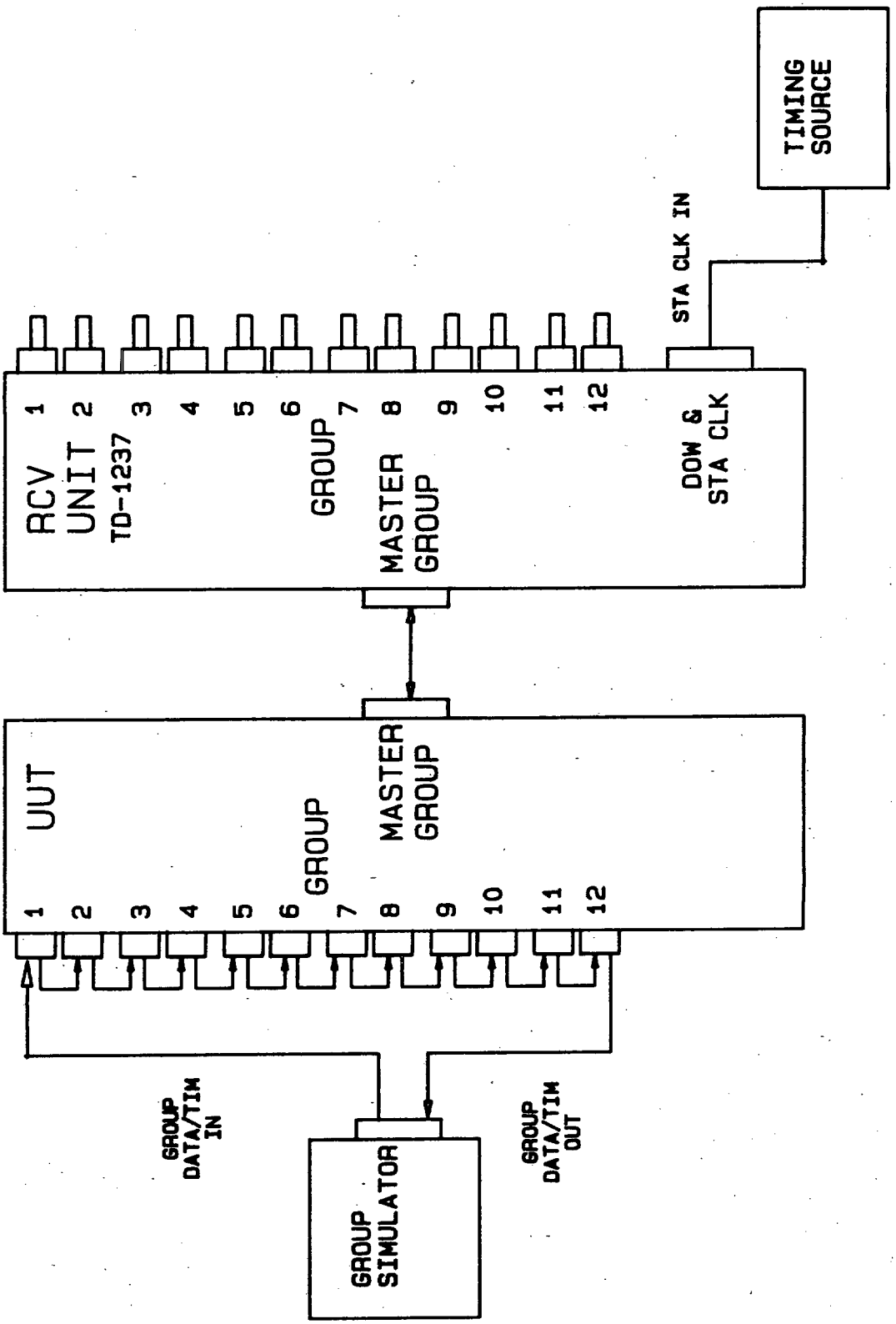


FIGURE 16. Timing modes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

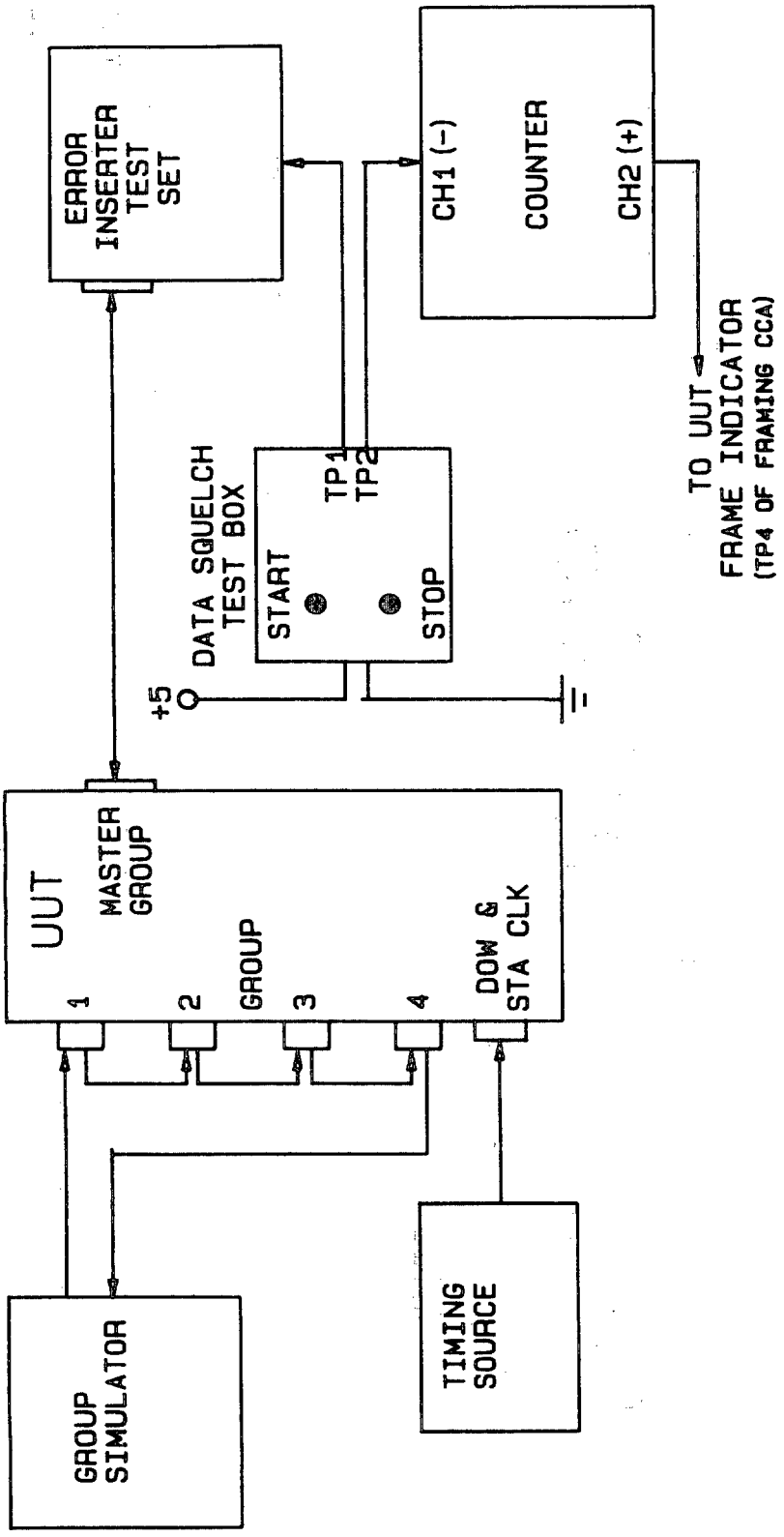


FIGURE 17. Framing.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

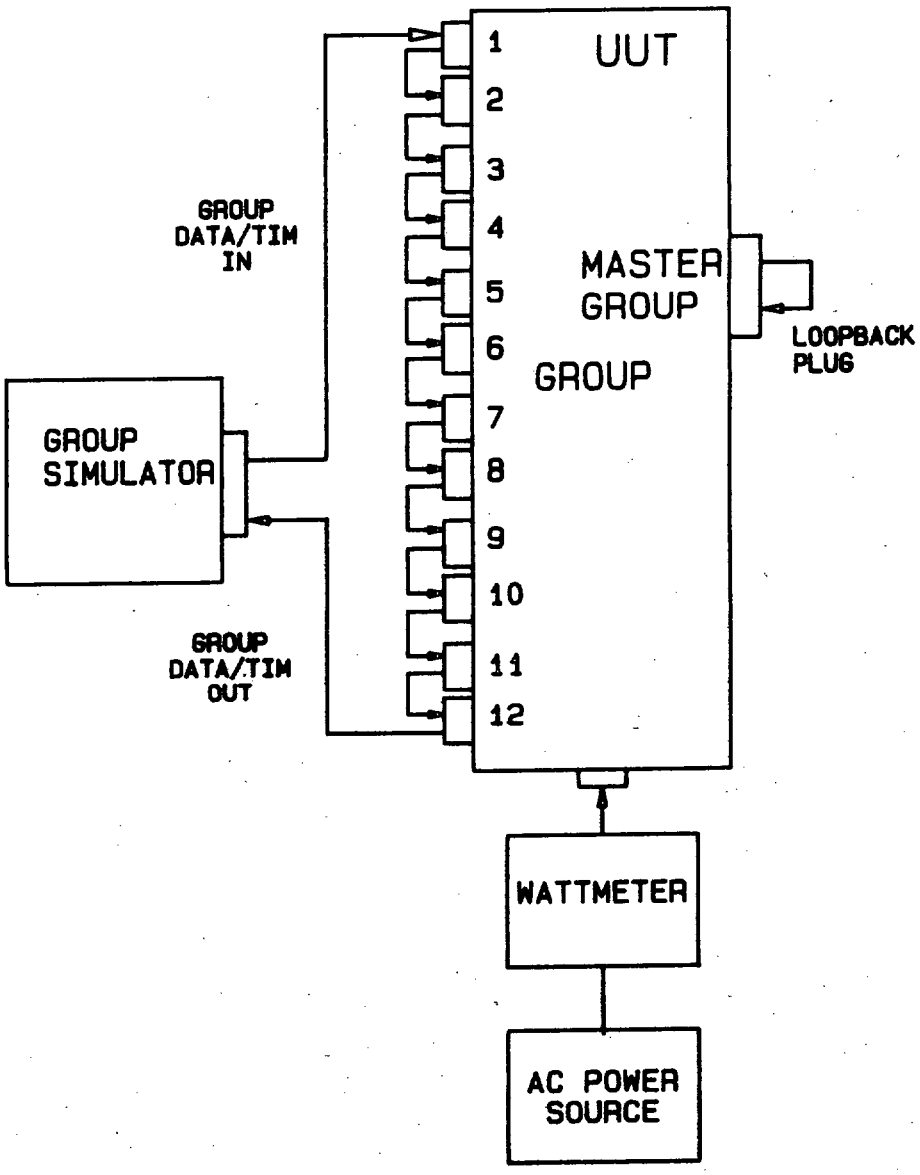


FIGURE 18. Prime power variation.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

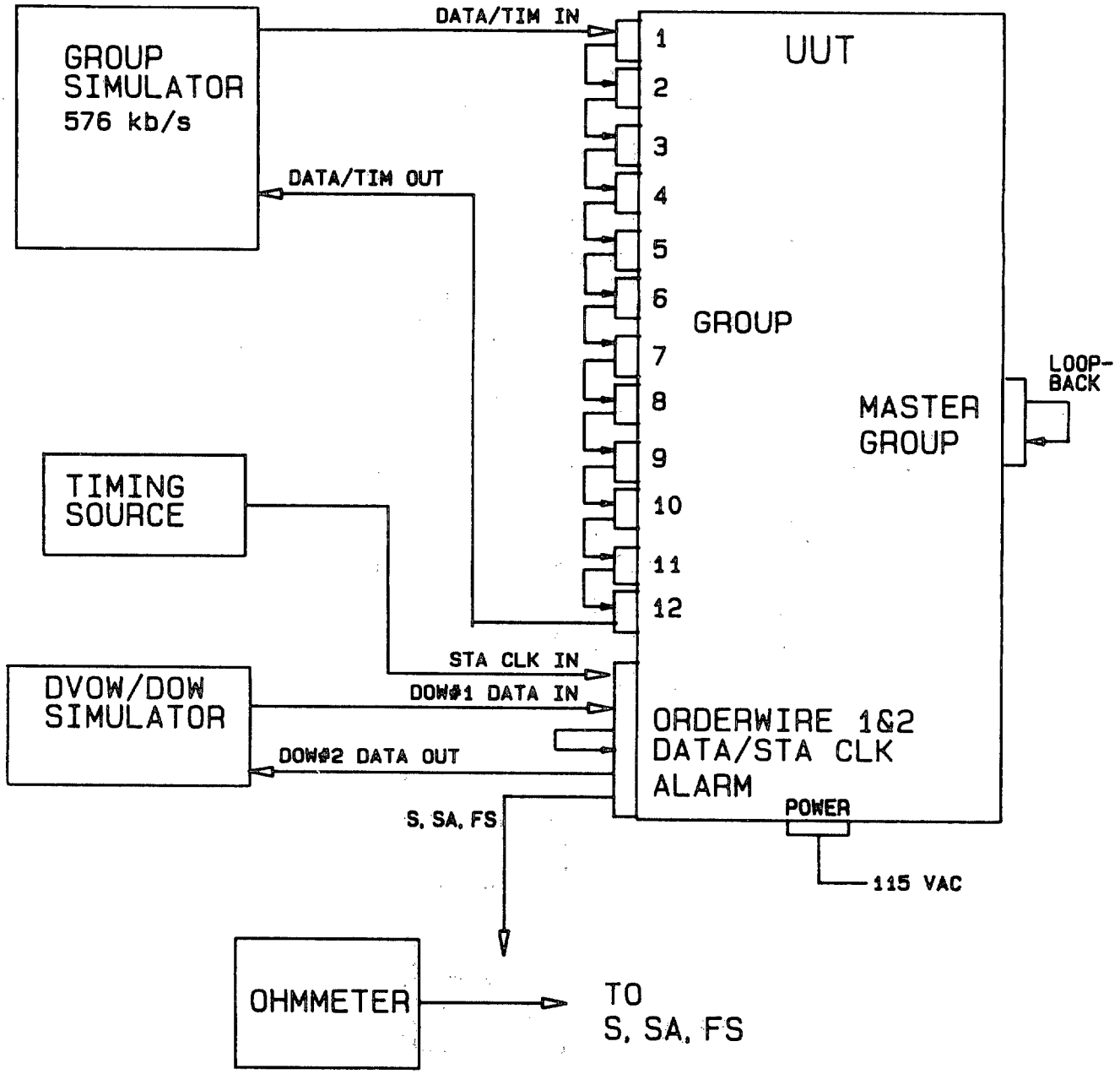


FIGURE 19. BITE and alarms operational check.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

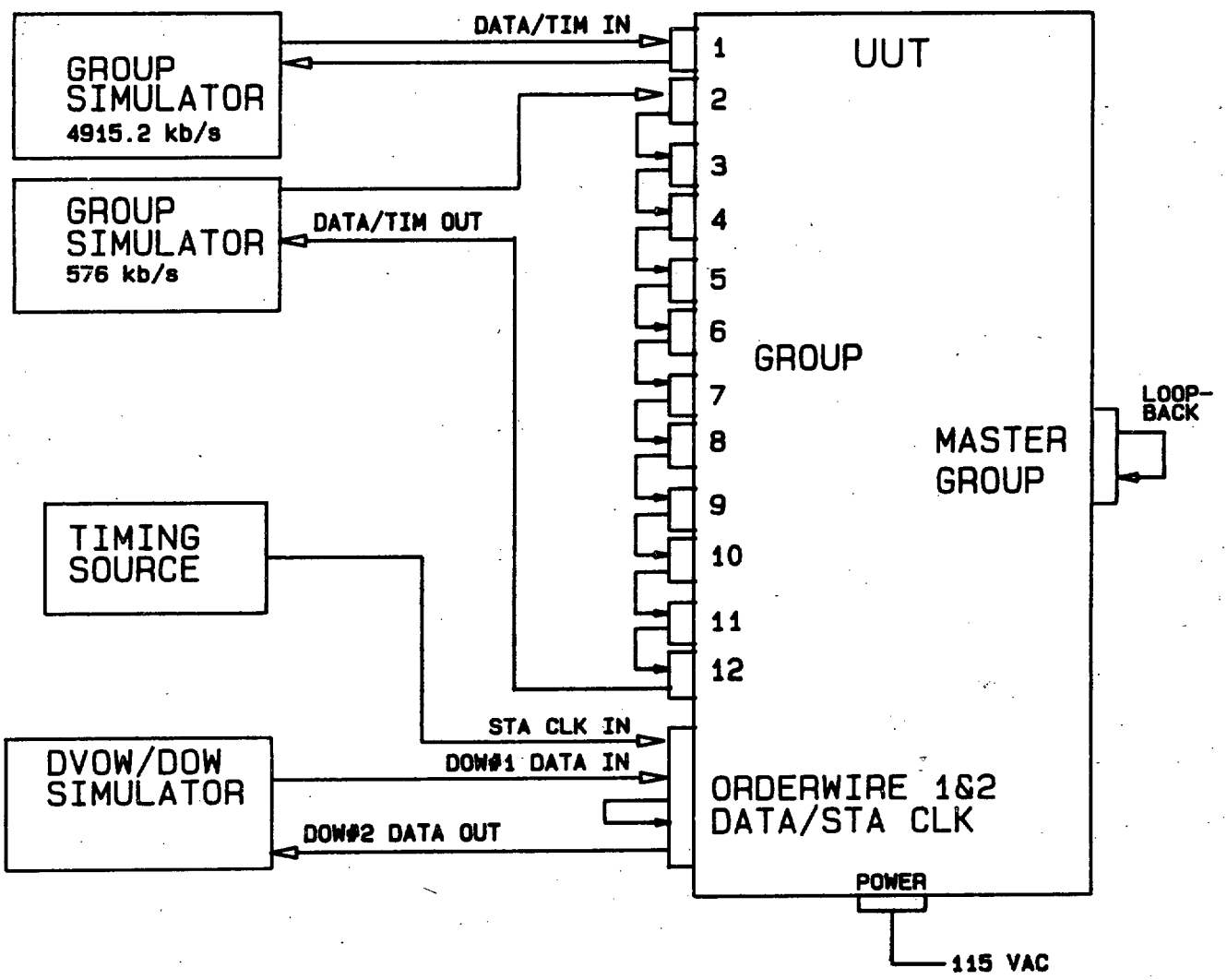


FIGURE 20. Operational check.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

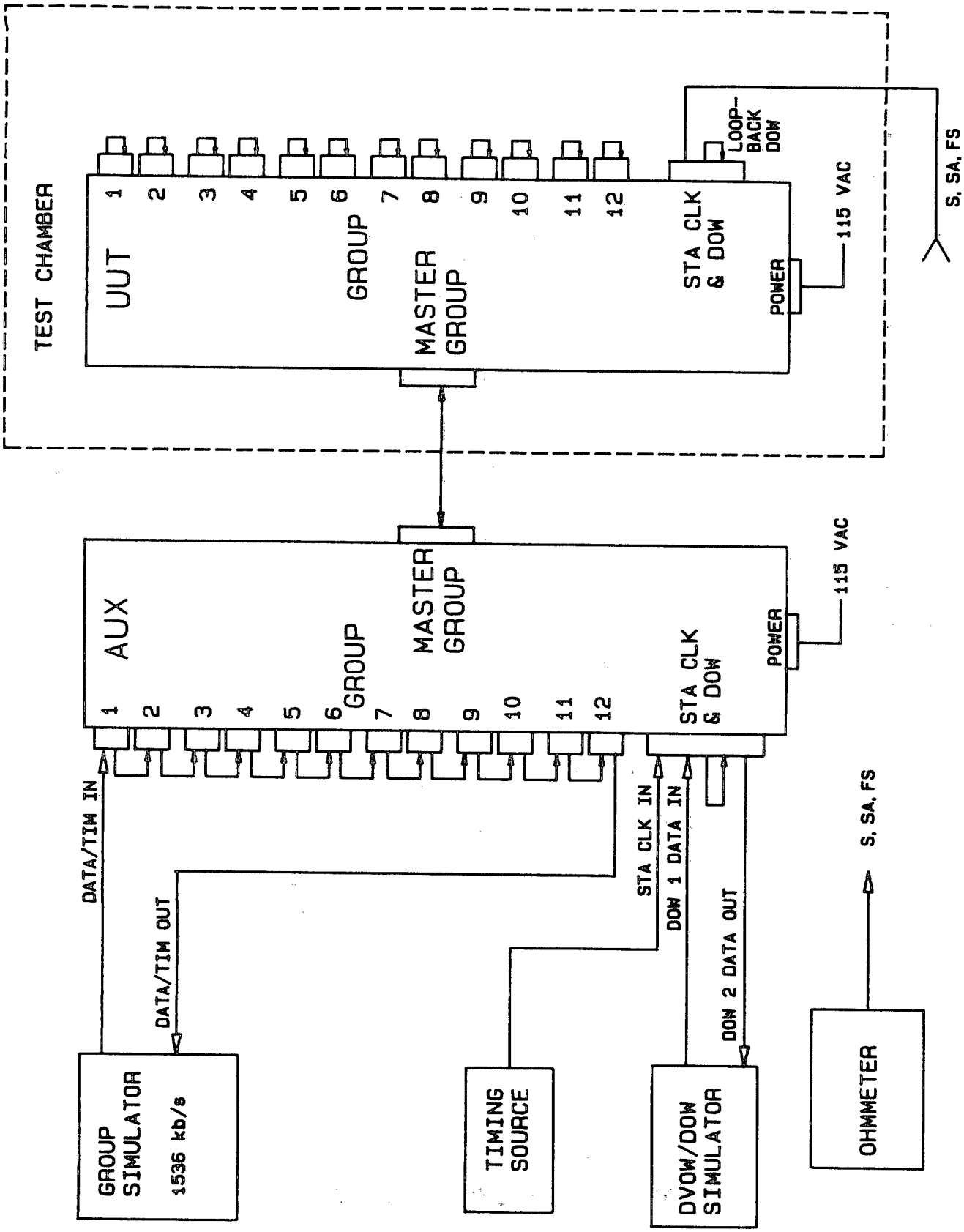


FIGURE 21. Environmental operational test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

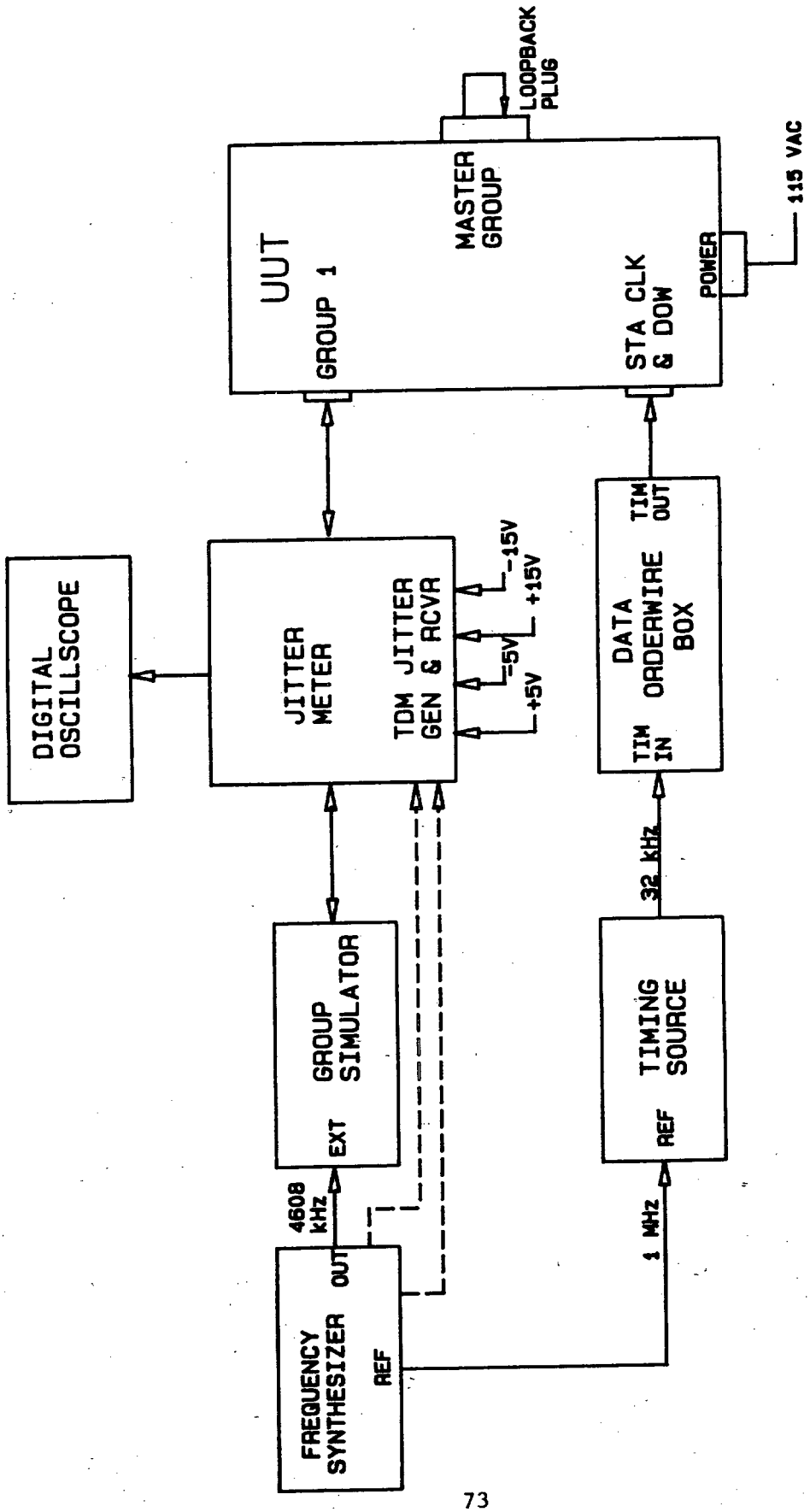


FIGURE 22. Jitter.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

☆U.S. GOVERNMENT PRINTING OFFICE: 1988 - 505-033/91735

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-49242A(CR)	Multiplexer, TD-1237(P)/G

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE. HISA-FM 149-85

Source: https://assist.dla.mil - Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.